SCALE WITHOUT SUPPORT: FRACTAL SIGNATURES ARE NOT DOCUMENT REPRESENTATIONS

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ABSTRACT

The Hurst and Hölder exponents that summarise language's fractal information structure separate human text from language-model output. Treating the exponents as an *estimand* and a document as a finite *support*, we show this separation is a property of the generating process in ensemble which does not descend to the individual document. On $223{,}834$ GAGLE documents the corpus-level gap survives full re-derivation of the human population and a determinism fix — 20 of 20 (domain, generator, exponent) cells separate with non-overlapping 95% intervals, with a second, differently biased estimator agreeing on direction in 10 of 10 Hölder cells — while on the *same* documents a pre-registered interleaved split-half check finds the flagship per-document Hölder estimate agrees with itself at Spearman–Brown $\rho = 0.063$, the best fractal quantity reaches 0.576, and 0 of 20 *candidates* clear a $\rho \geq 0.60$ bar declared in advance. At 400 tokens the estimate is dominated by estimator variance around a class-, domain- and temperature-conditional mean: it encodes *which population a document came from*, its document-specific component too small to survive our bar. Four predictions follow, all four hold (Table 1) — deleting all 54 fractal features from an 85-feature detector leaves AUROC at 0.9984 and TPR@1%FPR statistically unchanged ($0.9587 \to 0.9612$); the family recovers only about a sixth of the achievable signal once the population is balanced by construction; and it pays only when the population changes, $+0.0240$ AUROC on a matched RAID transfer, which we show is the population mean travelling. The result is a condition: treat a scaling estimator as a document representation only once its split-half reliability at the deployment length clears the estimator-variance floor.

1 INTRODUCTION

Language has a fractal information structure: its per-token complexity is self-similar across scales and long-range dependent, with a Hurst parameter near 0.7 (Alabdulmohsin et al., 2024). The observation extends to generated text. Alabdulmohsin & Steiner (2025) report that the Hölder and Hurst exponents of natural language occupy a narrow band while those of language-model output vary widely, and conclude that "fractal parameters might prove helpful in detecting LLM-generated text" — while declining to make the detection claim themselves, on the grounds that the exponents describe processes rather than documents. That caveat is usually read as a limitation of scope. We read it as a claim about the object being measured, and we test it. A scaling exponent is defined by a limit — $R(n)/S(n) \sim n^H$ as $n \to \infty$ — so every empirical use of one is an estimator on a finite realisation, and whether it resolves anything about that realisation, as opposed to the process that generated it, is a question about support. Related work (Appendix J) reports no reliability coefficient for a scaling estimator at document length.

Two things this result is not. It is not an attack on a claim nobody made: the step from a distribution-level separation to a per-document score is taken explicitly by the closest neighbour of this work (Tulchinskii et al., 2023), a construction now widespread as a detector (Guo et al., 2026; West et al., 2025; Liu et al., 2026; Mansouri et al., 2026; Teja et al., 2025; La Cava & Tagarelli, 2026) and as a characterisation of generated text (Du & Tanaka-Ishii, 2025; Yang et al., 2026). Nor

is it the foreseeable failure of a power-law fit over 1.6 decades, which would fail both exponents equally; we measure a near-order-of-magnitude split instead (0.576 against 0.063; both documented from primary sources, Appendix K).

Reconciling the population-level and document-level measurements. On $223{,}834$ GAGLE documents (Alabdulmohsin & Steiner, 2025) scored under Gemma-2B and Mistral-7B, 20 of 20 (domain, generator, exponent) comparisons separate with non-overlapping 95% intervals, a second and independently biased estimator agreeing on direction in 10 of 10 cells — and this holds under a full pipeline audit, not only the pass that first measured it (Appendix A). On those same documents, a reliability check registered before any detection experiment splits each token sequence into interleaved halves and correlates both exponents across the corpus: per-document $\widehat{S}$ agrees with itself at $\rho = 0.063$, $\widehat{H}$ reaches 0.576 against a bar of 0.60 fixed in advance, and zero of twenty pass. Writing the per-document estimate as $\widehat{H}_d = \mu(c_d, g_d, T_d) + \delta_d + \eta_d$, the first result says μ separates classes and the second says $\mathrm{Var}(\eta) \gg \mathrm{Var}(\delta)$ at $N \approx 400$ tokens: such a representation encodes population membership, not document content, and the consequences follow as deduction — the four predictions of Table 1, tested in §4–§6.

Contributions. What the paper is *about*, and what it also happens to establish independently, are separated below.

Primary — one claim, four predictions, one condition. The fractal signature is a property of the *generating process*, not of a document. **(C1)** An estimand/support formulation with the paired measurement that separates the two on identical documents (§3). **(C2)** Four predictions, each confirmed at an independent measurement point — three negative, one positive (Table 1). **(C3)** *Perturbation as a structural probe*: partitioning an edit battery by index versus realisation predicts the *sign* of the family's contribution in all six cases (§5). **(C4)** The admissibility condition this yields, costing one split and one correlation, independent of the downstream model.

Secondary — independent of the fractal question, and usable alone. **(S1)** *Scorer–attacker lineage inflates measured robustness*: a curvature detector's recall *rises* under paraphrase from its own scorer's lineage and collapses under any other (§7). **(S2)** *Perturbation-response features need a class-symmetric population*, or the drift feature becomes a label-leakage channel; ours is 100% machine-generated, and we specify the fix (§8).

The primary finding is not feature selection: it is *learner-invariant* (a shallow MLP reproduces it and fails harder) and *scorer-architecture-invariant* ($\rho = 0.063$ and 0.050 from a 2B Gemma and a 7B Mistral on the same documents), and §5 evaluates neural detectors end to end.

2 Background: Estimand and Support

2.1 Related work, and what is new here

Long-range correlation in text is an old observation (Ebeling & Pöschel, 1993; Altmann et al., 2012), formalised by Alabdulmohsin et al. (2024) at $H \approx 0.7$ and extended to model output by Alabdulmohsin & Steiner (2025), whose narrow- versus wide-band contrast is our starting point. Parallel lines measure self-similarity by correlation dimension (Du & Tanaka-Ishii, 2025), spectral scaling (Yang et al., 2026), and the persistent-homology dimension of a token-embedding cloud (Tulchinskii et al., 2023; Wei et al., 2025) — the closest neighbour, whose remedy for short text, enlarging the support, is our diagnosis run in reverse (full treatment: Appendix J). *That short-series scaling estimators are biased is established and is not our contribution* (Cannon et al., 1997; Millán et al., 2021; Grech & Pamuła, 2013; Weron, 2002). **We read that as corroboration, not a confound** — the estimator studied here is the one this literature deploys, so the reliability we measure is the reliability a practitioner obtains. Our step is the next one: what finite-support instability implies for treating such an estimate as a per-document *representation*, a reliability question in the psychometric sense (Pronk et al., 2022; Ellis, 2023). To our knowledge no prior work reports such a coefficient for a scaling estimator at document length.

Let q be a fixed scoring language model and $x = (x_1, \ldots, x_N)$ a document. The surprisal process and its integrated trajectory are

$$s_t \;=\; -\log q(x_t \mid x_{<t})\,, \qquad X_n \;=\; \sum_{t \le n} \frac{s_t - \mu_s}{\sigma_s}, \tag{1}$$

with μ_s, σ_s the document's own mean and standard deviation, so X is a mean-zero walk with unit-variance increments. **Hurst:** by rescaled-range analysis $\mathbb{E}\big[R(w)/\sigma(w)\big] \sim w^H$ for window length w, and $\widehat{H}$ is the OLS slope of $\log R/\sigma$ against $\log w$; $H = \frac{1}{2}$ is memoryless, $H \to 1$ strong persistence — in the limit, verbatim repetition. **Hölder:** for fixed $\varepsilon \ll 1$ and $p_\varepsilon(\tau) = \Pr\big[|X_{t+\tau} - X_t| \le \varepsilon\big]$, $p_\varepsilon(\tau) \sim \tau^{-S}$ and $\widehat{S}$ is again a log–log slope, smaller S meaning displacements stay inside ε for longer. Both are fit over $\tau \in \{8, 16, 32, 48, 64, 96, 128, 160, 192, 256, 320\}$ with $\varepsilon = 10^{-2}$, calibrated to Eq. 1's z-scoring (Appendix L); per document we drop the first 64 tokens as warm-up, clip to 464, and require 320 surviving tokens, dropping shorter documents rather than padding.

Before the algebra: this is the distinction between a town's climate and one day's thermometer reading (full analogy: §2.2). These equations define asymptotic properties of a process, but the realisation here has $N \approx 400$ usable tokens, across which the scale sweep spans a little over 1.6 decades. Two questions must be kept apart: **Q1 (ensemble)**, does H (resp. S) differ between the human and machine generating processes, answerable by pooling realisations (§3.1); and **Q2 (support)**, does $\widehat{H}(x)$ resolve anything about the individual x beyond the population it came from, a question about estimator variance at $N \approx 400$ (§3.2). They are logically independent, and a positive answer to Q1 is routinely — and, we argue, incorrectly — read as licensing a document-level feature. Decompose the per-document estimate for document d with class c_d and domain g_d as

$$\widehat{H}_d \;=\; \underbrace{\mu(c_d, g_d, T_d)}_{\text{population term}} + \underbrace{\delta_d}_{\text{document term}} + \underbrace{\eta_d}_{\text{estimator error}}, \qquad \mathbb{E}[\eta_d] = 0. \tag{2}$$

Q1 asks whether μ separates over c; Q2 asks about $\mathrm{Var}(\delta)/\big(\mathrm{Var}(\delta) + \mathrm{Var}(\eta)\big)$. Both μ and δ are shared between a document's halves while η is not, so μ *enters the numerator*: pooled over a corpus whose documents differ in class and domain, the half–half correlation estimates $\mathrm{Var}(\mu{+}\delta)/(\mathrm{Var}(\mu{+}\delta) + \mathrm{Var}(\eta))$, reliability with respect to population *plus* document, so every coefficient below is an **upper bound** on the quantity Eq. 2 names (Ellis, 2023). **The bound is measured, not left abstract**: demeaning each half by its own (c, g) stratum before correlating gives ρ_{within} directly (Appendix B) — 0.552 for `gemma_H` and 0.520 for `mistral_H` against $\rho_{\text{pooled}} = 0.576/0.547$, and $0.060/0.046$ against $0.063/0.050$ for $\widehat{S}$: population structure explains under 4% of single-half variance for $\widehat{H}$ and under 0.2% for $\widehat{S}$. **This closes an escape route rather than opening one**: $\widehat{S}$'s near-zero reliability is document-level estimation noise, present almost unchanged once the population term is removed by construction.

2.2 The same idea without the algebra, and the four predictions it makes

Two neighbouring towns differ by half a degree in mean annual temperature. The difference is real, and a year of daily readings from each resolves it tightly. Now take a *single* reading from a thermometer whose error is $\pm 5^\circ$ and ask which town it came from. Nothing about the towns has changed; the reading simply cannot separate them, the instrument's noise dwarfing the quantity of interest at that sample size. **The climate is $\mu(c_d, g_d, T_d)$, the individual house is δ_d, the thermometer's error is η_d** — the whole of Eq. 2, and the algebra is done. The story does not merely illustrate the result; it *predicts* four things, each a separate chance to be wrong. **(P1)** Knowing which country you are in, a second thermometer adds nothing — deleting the fractal features from a detector that already has perplexity and style should cost nothing. **(P2)** A noisy needle twitches constantly and says nothing — a greedy learner should *use* these columns heavily and *gain* nothing. **(P3)** The sharpest: ask it to distinguish *two houses on the same street* and little should be left, the climate term being common to both. **(P4)** And the story must say where the thermometer is still worth carrying — *move to a different country* and it informs again, because the climate is what changed. Table 1 maps the four tests to the four sections that follow.

One distinction the story hides: reliability and bias are separate, and this estimator fails on both. Per-document $\widehat{S}$ is biased low at 400 tokens by ≈ 0.15 against the pooled estimate, which is why corpus-level numbers below re-pool raw trajectories rather than averaging point estimates.

Table 1: The paper in one table: each row is a prediction the thermometer story makes, the test, and the outcome. P1–P3 negative, P4 positive.

	In the story	The test	Result
P1	A 2nd thermometer adds nothing, knowing the country	Delete all 54 fractal cols. (§4.1)	AUROC unchanged 0.9984; recall $0.9587 \to 0.9612$, CIs overlap
P2	The needle twitches constantly, says nothing	Split gain vs. permutation importance (§4.1)	5.52% of gain, 1.7×10^{-4} AUROC held out
P3	**Two houses, same street**	8-way prompt type, machine rows only (§4.2)	0.198 macro-F1 vs. 0.125 floor; all 85 features reach 0.582
P4	Move country, thermometer useful again	GAGLE→RAID, population-matched (§6)	$+0.0240$ AUROC $[0.0222, 0.0259]$, excludes zero

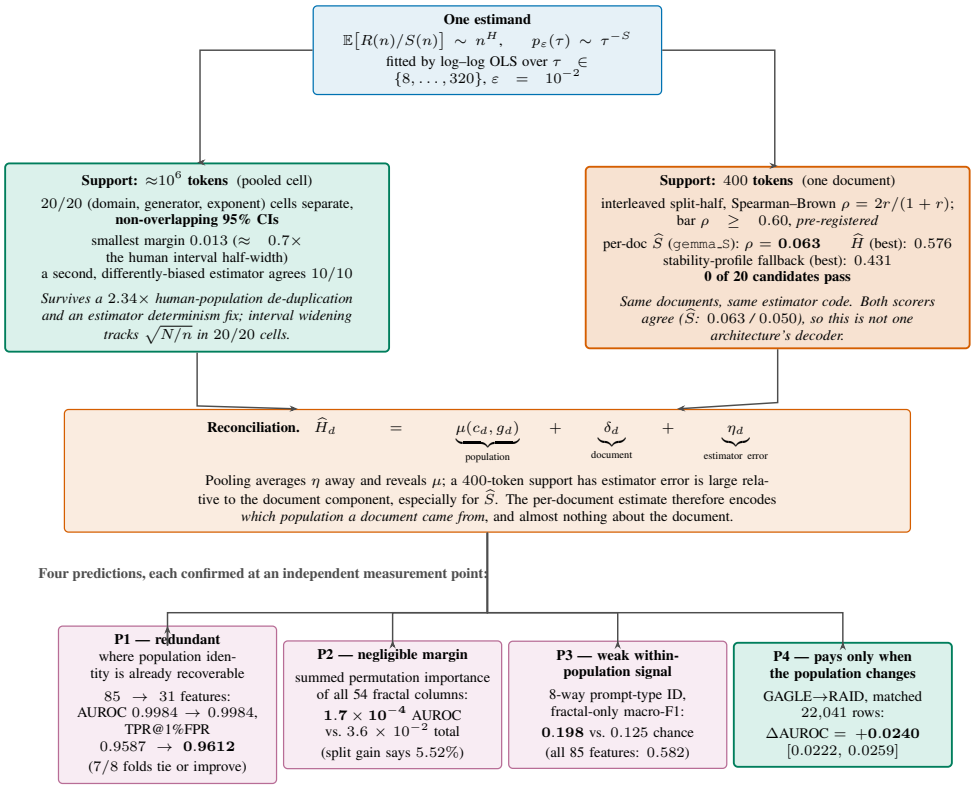


Figure 1: One estimand, two supports. *Left:* pooled, the exponents separate in $20/20$ cells with non-overlapping intervals. *Centre:* the same documents at 400-token support give $\rho = 0.063$ for $\widehat{S}$ and $0/20$ features clearing $\rho \geq 0.60$. *Right:* the four predictions Eq. 2 makes, each confirmed independently.

3 The Dissociation, Measured

All measurements are on GAGLE (Alabdulmohsin & Steiner, 2025): five domains × two generators (Gemma-2B, Mistral-7B) × three temperatures × eight prompt types, plus the human source articles. After the token-window gate the store holds 223,834 documents — 112,402 Gemma-2B, 106,432 Mistral-7B, 5,000 human — at a human prevalence of 2.234%. Each carries 85 features in three families, enumerated in full in Appendix I: **A (fractal,** 54), both exponents with fit quality, uncertainty, per-scale residuals and windowed profiles; **B (token-statistical,** 17), log-perplexity moments, rank buckets (Gehrmann et al., 2019) and entropy; **C (discourse/style,** 14), length, repetition and dispersion statistics on a ratio-derived word budget so they cannot degenerate into length proxies. Family-B features are written only above an $r_{\text{align}} \geq 0.98$ gate, GAGLE's precomputed scores recording neither checkpoint nor tokenizer.

3.1 Q1: THE ENSEMBLE GAP SURVIVES FULL RE-DERIVATION

Before treating the corpus-level separation as established, we audited it against two potential confounds in the corpus and estimator pipeline, and report the fully audited result below rather than the first version measured. First, population: GAGLE's split label was treated as part of an article's identity, so the same article recurred as several "different" human documents; collapsing these duplicates takes the human N from $\approx 11{,}700$ to $5{,}000$. Second, estimator determinism: the Hölder estimator sampled window starts rather than enumerating them, so every stored per-document $\widehat{S}$ carried Monte-Carlo noise; enumerating them removes it. Both audits are resolved, and every S-based interval below is recomputed on one estimator vintage; the smallest margin anywhere is `wikipedia` S under Gemma-2B, a gap of 0.013, about 0.7 of the human interval's half-width (Table 2, Appendix A). **A second, differently biased estimator agrees**: bootstrapping the mean of per-document point estimates separates every cell in the same direction, $10/10$ on S (two further checks and a replication note: Appendix A).

3.2 Q2: THE PRE-REGISTERED RELIABILITY GATE, AND ITS FAILURE

The check was specified, with its threshold, before any detection experiment was run. Each document's token sequence is split into *interleaved* odd/even halves — not first-half/second-half, which would confound estimator unreliability with genuine stylistic drift between a document's opening and its close — both exponents are refit independently on each half, and the half-length correlation is Spearman–Brown corrected, $\rho = 2r/(1+r)$, against a $\rho \geq 0.60$ bar declared in advance.

The interleaving assumption, and its scope. Both estimators are functionals of sequential geometry, so it is fair to ask whether decimating the sequence changes the object measured. Under this paper's own premise it does not: for a process exactly self-similar with stationary increments — the premise that makes H and S well defined at all — decimation by two *is* the dilation the scaling law is invariant under, so each half carries the same exponent up to a known constant. **What the objection does reach is the gap between exact scaling and real text**, outside that argument, so we tested the sensitivity of ρ to the split construction directly rather than leaving it open.

The split-construction test. Eight constructions were run end to end on the full corpus (enumerated, with a ninth measured but excluded as invalid, in Appendix B). **Reliability declines smoothly and monotonically as the block grows** — `gemma_H`: $0.576 \to 0.562 \to 0.549 \to 0.534 \to 0.501$ for block sizes $1, 4, 8, 16, 32$ — the opposite of what the sequence-geometry objection predicts: if decimation damaged the estimand, coarser blocks should *recover* reliability, and instead single-token interleaving is the best-performing of the eight, tied by the full sweep. Only `contiguous` falls outside the band every other construction shares, the pre-registered, *expected* signature of the drift confound the design exists to avoid. **Interleaving is not the outlier that inflates ρ; it is the construction that damages the sequence least.**

Zero of twenty candidates pass, and the result separates into two qualitatively different findings (full table, Appendix B). $\widehat{H}$ has modest but genuine reliability falling just short of the bar ($\rho = 0.576$ for `gemma_H`, 0.547 for `mistral_H`); $\widehat{S}$ has essentially none (0.063 and 0.050). **Read $\rho = 0.063$ concretely: at $N \approx 400$ tokens (a few short paragraphs), measure one document's Hölder exponent on its odd tokens and again on its even tokens, and the two answers agree about as well as two documents chosen at random.** That is not a weak effect: no plausible correction moves 0.06 to 0.60, and the natural rescue fails with it — the pre-registered windowed-stability fallback tops out at 0.431 and bottoms out at 0.003, a shorter instrument rather than more curve (Appendix B). **The two exponents fail differently and we do not treat them as one:** $\widehat{S}$ is unreliable here, while $\widehat{H}$ is moderately reliable and below the bar by more than sampling error — at $n = 221{,}904$ its Fisher-z 95% interval is $[0.573, 0.580]$, so $\widehat{H}$ misses by construction, not by noise. The result is architecture-independent across a 2B Gemma and a 7B Mistral, taken *after* the estimator's earlier non-determinism was fixed and the corpus regenerated.

Estimator ablation: DFA and local Whittle. The verdict is mixed and we report it as mixed. Refit on the identical halves, local Whittle clears the bar for both scorers ($\rho = 0.677/0.642$); DFA clears it for Gemma only, marginally, and misses for Mistral. R/S and DFA correlate at $0.87/0.86$

(the same exponent, less noisily estimated); R/S and Whittle correlate at only 0.76/0.75, so part of Whittle's headroom is not shared with the exponent $\widehat{H}$ is defined against (full numbers and a third, geometric estimator: Appendix B). **Neither estimator touches** $\widehat{S}$, whose reliability stands exactly where §3.2 already put it.

Estimator bias against ground truth, by support. Reliability and bias are separate properties and this estimator is compromised on both: validated against synthetic fractional Brownian motion of known exponent, at document support $\widehat{S}$ is biased -0.26 to -0.48, and more bootstrap draws at the same fixed support do not close the gap (full ladder and pooling-mode bias: Appendix B). Since every machine cell sits above its matched human cell in the pooling mode actually used, this bias is conservative for the ensemble gap: understated, not manufactured.

Reliability as a function of support, and where it crosses the bar. The bound above says how much of ρ_{pooled} is population; it does not say how much *support* would fix the rest. We calibrate an estimator noise floor against synthetic fractional Gaussian noise at GAGLE's own observed $H \in [0.60, 0.72]$ band and compose $\rho_{\text{SB}}(N) = 2r(N)/(1 + r(N))$ across $N \in \{100, \ldots, 12{,}800\}$ (Figure 3, Appendix B). $\widehat{H}$ crosses the bar at $N^\star = 800$ — already $\rho_{\text{SB}} = 0.49$/0.46 at $N = 400$, close to GAGLE's own support — while $\widehat{S}$ needs $N^\star \geq 6{,}400$, sixteen times what any document here supplies. A companion pass over $5{,}000$ full-length human documents confirms the floor is not an underestimate of real-text scatter (median full-document $\widehat{H} = 0.629$, inside the band). **That check is $\widehat{H}$-only**: for $\widehat{S}$ it predicts 0.229/0.197 at $N = 400$ against 0.063/0.050 measured, so $\widehat{S}$'s $N^\star$ is a *lower* bound. **The asymmetry between the two exponents is now quantitative, not just qualitative**: $\widehat{H}$'s shortfall resolves at roughly twice GAGLE's token budget; $\widehat{S}$'s does not resolve within an order of magnitude of it.

The gate's designed output, a whitelist of surviving fractal features, was emitted empty; the detection stage trains on all 54 from a manifest declared in advance instead, the bar stays at 0.60, and the features are carried forward so the ablation below can price them independently.

4 What the Document-Level Estimate Represents

That the per-document estimate is dominated by estimator error at this support is a statement about an instrument, and on its own proves nothing about detection — an unreliable quantity could still be useful by accident. This section and the two that follow close that gap by running the four tests of Table 1 in order: P1 and P2 are consistency checks, **P3 is the strongest of them** because it holds the climate term fixed by construction and so isolates δ, and P4 is the positive control that makes the account falsifiable rather than merely pessimistic.

The instrument. The learner is deliberately unremarkable and held fixed across every comparison: a class-weighted gradient-boosted tree ensemble, splits frozen before training and group-disjoint by seed article. Eight folds cross five confound configurations, arms and learners (304 cells, $2{,}000$-fold bootstrap intervals); the headline metric is recall at fixed low FPR, not AUROC. Two controls are load-bearing (Appendix C): a label-shuffle harness landing systematically *below* chance, and an explicit pricing of the length confound, a two-feature `length_only` arm reaching AUROC 0.8635 at TPR@1%FPR 0.0477.

4.1 P1 and P2: redundant in-distribution, worthless at the margin

In the story: a second thermometer, handed to someone who already knows which country they are standing in. Deleting all 54 features leaves AUROC unchanged to four decimals and *improves* recall at 1% FPR from 0.9587 to 0.9612: a 31-feature model matches an 85-feature one (full arm grid, Appendix C). **This outcome is predicted by §3.2, not merely consistent with it.** Without the reliability measurement in hand, a family separating human from machine text in 20 of 20 cells yet contributing nothing to a classifier on the same corpus reads as a modelling failure and invites an architecture search. It is neither: $\rho = 0.063$ says the estimate is a population-conditional mean plus noise, and perplexity and style already identify the population, so the fractal columns arrive carrying information the model has, wrapped in variance it does not want. A fractal-only model reaches

AUROC 0.6629 above chance, but the family is neither necessary nor additive: `fractal+stat` sits *below* `stat_only`, `all_minus_fractal` ties or wins on 7 of 8 frozen folds, and a shallow MLP reproduces the pattern (Appendix C).

P2 is the fingerprint. Split gain over all 85 inputs credits the 54 fractal columns with 5.52% of the total, and the model splits on every one at least 125 times. Permutation importance on the held-out fold tells the opposite story: the summed AUROC drop across all 54 is $\mathbf{1.7 \times 10^{-4}}$ against 1.54×10^{-2} and 2.04×10^{-2} for the other two families — 0.47% of the total against a 5.52% gain share, with 35 of 54 zero or negative (full breakdown, Appendix D). **Used constantly, worth nothing.**

4.2 P3 — THE STRONGEST WITHIN-POPULATION TEST: HOLD THE POPULATION FIXED

In the story: two houses on the same street. P1 and P2 show the representation is redundant and marginally worthless; neither shows *why*, since both are measured where the population term is available. Eight-way prompt-type identification, run on machine-generated GAGLE rows only, removes that escape route: GAGLE is a complete 240-cell factorial (2 generators $\times 3$ temperatures $\times 5$ domains $\times 8$ prompt types), so prompt type is crossed with every population factor *by the corpus's own design*. μ **varies across documents but is balanced across the labels being separated**, contributing variance without contributing signal — *of the tests here, this is the one best placed to have falsified the account.* **The task is solvable, and the fractal representation recovers little of it.** Chance is macro-F1 0.125. All 85 features reach 0.5816 [0.5747, 0.5885], so these eight classes genuinely differ document by document; the 54 fractal features alone reach 0.1981 [0.1919, 0.2043] — **about a sixth of the distance** from guessing to what the corpus supports, from a family with $1.7\times$ as many columns as the incumbents. **We do not stop at asserting the balance survives the gates; we measured it**, with three checks that close the residual-confound route directly (full numbers, Table 20, Appendix H.1): a population-only control (generator, domain, temperature, no prompt type) scores *below* chance (0.0594, floor 0.125); the factorial survives the token-window and complete-case gates with a maximum cell deviation of 0.22 percentage points; and removing each fractal column's stratum mean before refitting leaves fractal-only macro-F1 *unchanged, its confidence interval overlapping* ($0.1981 \rightarrow 0.2046$) rather than falling toward chance. **Together these rule out the residual-population-confound explanation rather than merely bound it**: the signal is genuine document-level structure, not population leakage. This is a within-population discriminative test, not a direct estimate of $\mathrm{Var}(\delta)$, and does not reference the split-half gate — so $\rho = 0.063$ should be read as a property of the estimand–support relationship, not an artifact of how we split tokens. The number most exposed by this analysis is unchanged: 0.5816, not 0.1981.

Two further within-corpus results separate the same two terms (full folds and numbers: Appendix C). *Substitution*, a question about δ: the fractal arm loses on **all eight of eight** folds against `stat_only`. *Generator identity*, a question about μ, is easy by contrast: AUROC 0.9943, same corpus, same features. **Same features, same corpus, same classifier: near-chance when asked about the document, near-perfect when asked about the population.** Where the detector actually fails is not a fractal story either — a tail-ranking failure on `billsum` LODO driven by prompt type, whose largest shifts are style features.

5 THE STRUCTURAL PROBE: DIRECT EVIDENCE FOR THE MECHANISM

In the story: jostling the thermometer while you read it, versus repainting the house it stands in. **This section is a targeted diagnostic of what the fractal features are keyed to, not an adversarial-robustness benchmark; its Class I result is the paper's strongest evidence for the proposed mechanism.** We ask what structure the representation reads, not how much recall an attack removes, using two kinds of edit that destroy different things. The scaling estimators of §2 are functionals of one structure only, the ordered token index: indifferent to *which* words appear, sensitive only to their arrival order. The bank splits on that axis: **Class I (order)** — `sentence_reorder`, `typo_noise` — scrambles *where* tokens sit, not *which* words they are; **Class II (content)** — four paraphrase/style/summarize operators — swaps *which* words appear at fixed coarse-scale order (Appendix E). A document-level property should need Class I most; population identity plus estimator noise should make Class I actively *harmful* instead.

The battery applies six operators to GAGLE machine text, yielding $39{,}810$ (document, variant) pairs, each passing a semantic-preservation guard (full grid, detector list and definitions: Appendix E). The tabular detector (Table 15) stays above 0.92 on every Class II operator and collapses to 0.2670 on `sentence_reorder`, while Binoculars (Hans et al., 2024) degrades roughly uniformly and a fine-tuned RoBERTa is unmoved.

Retraining the fractal-free 31-feature model on the same edited population, the partition predicts the sign in six of six cases: full model ahead by 0.94–2.10 points on all four Class II operators, and *deleting* the fractal family *raises* recall by 1.96 and 8.76 points on `sentence_reorder`/`typo_noise` — once order is scrambled, fractal features vote against the right answer, and dropping them recovers it.

This Class I reversal is why this section is diagnostic, not a secondary robustness check. `sentence_reorder` and `typo_noise` are the only two operators that attack the *support* rather than the *content*, preserving the lexical statistics every other feature reads and perturbing only the index Family A claims to measure. A representation that genuinely encoded that structure would degrade under them; instead the contribution *flips sign* precisely and only here. **A feature family cannot be a measurement of sequence structure and also be actively harmful when, and only when, sequence structure is perturbed.** This is what $\mathrm{Var}(\eta) \gg \mathrm{Var}(\delta)$ predicts, and it isolates δ without relying on the split-half gate (mechanism and qualifications: Appendix E).

6 P4 — Where Ensemble Structure Does Transfer, and Why That Is Not a Rebuttal

In the story: you move to a different country, and the same $\pm 5^\circ$ thermometer starts telling you something again. **This subsection addresses the one result that, at first reading, appears to contradict the paper's central claim.** Three sections have just shown a feature family to be redundant, marginally worthless and near-chance within a population; here the same family buys a statistically significant $+0.0240$ AUROC. If that were evidence the features carry document information after all, the argument would collapse. It is not, and the reason is the thermometer that has not changed while the climate around it has.

We test P4 on RAID (Dugan et al., 2024), 11 generators and 8 domains re-scored through the identical pipeline and used only as an external test, with *no* retraining. Crossing corpora costs 0.195 AUROC ($0.9984 \to 0.8032$) and 0.731 recall at 1% FPR ($0.9587 \to 0.2277$) — the largest degradation anywhere in this evaluation — and the fractal family's advantage survives on a composition-matched population (Table 18, Appendix H): small, $+0.0240$ AUROC on a model that is itself badly degraded, and real.

Testing the population-mean explanation directly

What moved is μ; δ is as absent here as in §4.2. Crossing to RAID changes the populations, and the fractal columns — carrying a coarse population mean that still correlates with generator identity in the new corpus — are left holding the one thing that survived. Tested directly: **replacing** each document's 54 fractal columns with its $(\text{generator}, \text{domain})$ group mean *recovers more than the entire* gap (AUROC 0.8149, 149% of it), and **subtracting** it retains 59% (AUROC 0.7934) — population information does the majority of the work, with a real document-level residual (full arms, Appendix H). **The honest reading is not "fractal features are robust" but "the ensemble term is the part that travels", now shown rather than only argued.**

What would have to be true for this to be a rebuttal is not true. A document-level source for the $+0.0240$ would require these features to carry document-level content somewhere; on the one task holding the population fixed they reach 0.198 against a 0.125 floor. Three facts bound it the same way: the gain appears only after the model has already lost 0.195 AUROC crossing corpora; `fractal_only` does not transfer; and it is population-level by construction, generator identity within GAGLE being recoverable at AUROC 0.9943 from features that cannot separate two machine-written articles. **Read as a recommendation, P4 says keep these columns when you expect the population to change, and never expect them to describe the document in front of you.** This is a *complete-case* result covering 32% of RAID's rows (excluded majority: Appendix L).

7 A Secondary Finding: Scorer–Attacker Lineage Inflates Measured Robustness

This section is independent of everything above. It uses the same edit battery but is not about fractal features, and nothing in §3–§6 depends on it; we report it separately because it is reproducible and will silently affect any robustness evaluation built the way ours was.

Fast-DetectGPT's recall *rises* under every Class II operator, from a 0.5104 baseline to 0.63–0.86 — at face value, paraphrasing making machine text easier to detect, the opposite of this literature's central finding (Krishna et al., 2023; Sadasivan et al., 2025). It is neither a floor artifact (far above our 15% no-signal screen) nor robustness: Fast-DetectGPT scores curvature under Gemma-2B, and these paraphrases were generated by a model in Gemma's own lineage, so the edited text is maximally predictable *to the scorer* and the attack raises the detector's key signal instead of defeating it. Regenerating the same paraphrase outside that lineage collapses recall to 1.2–4.5% across a capability ladder of five external models on $5{,}000$ documents, and the reversal is stable, the tabular detector degrading broadly monotonically with attacker size ($72.6\% \to 36.5\%$) and Binoculars tracing the cleanest curve ($70.8\% \to 35.1\%$; Appendix G). **Scorer–attacker lineage, not detector design, determines whether a curvature-based detector appears robust to paraphrase. The check is cheap: regenerate one attack type with a model outside the scorer's lineage and compare.**

8 Discussion, Limitations, and Conclusion

A claim about measurement, not about the existence of fractal structure. Nothing here contradicts language's fractal information structure, or model output's departure from it. This paper adds the second half: an exponent defined by an asymptotic law, estimated on a 400-token realisation across 1.6 decades, resolves the process far better than the realisation, and the whole curve does not rescue it. Support from outside this estimator family: Wei et al. (2025) find a topological scaling dimension on *embeddings* degrading at short length for the same reason, and Yang et al. (2026) report a $5/3$ spectrum disrupted by randomization of token order. **We also tested this directly** on a representation with no fractal machinery at all — overlapping-window MiniLM embeddings reduced to persistent-homology loop statistics, at zero content overlap between halves: no window geometry tried clears $\widehat{H}$'s own reliability (best genuine value 0.41; full pilot, Appendix L), evidence for the generality of the mechanism, not only for this estimator family.

For deployment the reading is regime-dependent: drop the family within a known corpus (redundant at best, costing 8.76 points of recall at worst); keep it for a detector that must generalise ($+0.0240$ AUROC). **No detector evaluated here supports consequential accusations against individuals**: the best loses 73 points of recall crossing corpora.

Limitations. Appendix L states each with its numbers, and one is a contribution in its own right: *perturbation-response fusion requires a class-symmetric edit battery*, since on a $9{,}997$-of-$9{,}997$ machine-generated population a drift feature exists only for the positive class, and a classifier given it learns population membership rather than robustness. The sensitivity of ρ to the split construction and the token support at which it clears the bar, previously open, are now measured (§3.2); matched-length bands, RAID complete-case composition, deployment-prevalence recalibration and a second representation family are in Appendix L. Beyond these: no architectural account of the exponents; no coverage of human revision; a single scorer pair, one corpus family, English only; an uncorrected 65-comparison run; and two feature-extraction defects found during review, after which every store was rebuilt, every number post-correction *with one marked exception* (Table 15).

Conclusion. The fractal signature of machine-generated text is real, and at this support it is not a document representation: 20 of 20 cells separate in ensemble, 0 of 20 split-half candidates clear a pre-registered bar at 400 tokens, and all four predictions of Table 1 hold. We propose as a measurement-validity criterion, not a theorem: a scaling statistic should not be read as a stable document-level measurement without a reliability coefficient at the support on which it is used.

AI USE STATEMENT

In this work, we used generative AI tools for none of the tasks carrying a required disclosure: no part of the research ideation, hypothesis formulation, experimental design, theoretical framework, or interpretation of results was generated by an AI system, no synthetic data sets were produced by generative AI for use as evidence, and no mathematical or empirical claim originates from an AI system. Generative AI was used for tasks in the recommended-disclosure category only: drafting and copy-editing prose in this manuscript, literature search and retrieval of candidate related work, and assistance with LaTeX formatting and figure code. Every AI-assisted passage was checked against the underlying artifact it describes — each numeric claim in this paper carries a source comment in the LaTeX naming the file it was read from, and those were verified by the authors against the artifacts on disk rather than accepted from a generated draft. All retrieved references were opened and their venue, author list and content confirmed by the authors before citation. We take responsibility for the final content of this work, including all text, claims and artifacts.

ETHICS STATEMENT

Detection of machine-generated text is dual-use, and this paper's structural analysis is more dual-use than a detector would be: the operator partition of §5 tells a defender which feature family to drop and tells an attacker which edit class to prefer. We judge publication net positive on three grounds. First, the vulnerability reported — a published feature family degrading under sentence reordering and character noise — concerns an existing, already-public construction rather than a novel attack technique, and both operators are trivially available. Second, the paper's operational recommendation is simplification, dropping a costly and unreliable component, not disclosure of an exploit. Third, the scorer–lineage finding (§7) corrects an evaluation error that currently causes robustness to be *overstated*, which is a defender-side benefit. Against that: false accusation is the dominant harm in the settings where these systems are used, which is why every headline number here is recall at 1% or 5% false-positive rate rather than AUROC, and why §8 states plainly that no detector evaluated here supports consequential individual accusations. We did not measure per-writer-population false-positive rates — in particular for non-native writers, where detector bias is documented — and we flag that absence rather than let low aggregate FPR imply low FPR for every group. All data are existing public research corpora (GAGLE, RAID); no new human-subjects data was collected, and no human annotation was commissioned.

REPRODUCIBILITY STATEMENT

Every number in this paper is reproducible from artifacts already on disk without re-running any generation or scoring pass. Splits are frozen and versioned (`data/splits_v1.json`) and reused unchanged by the robustness and transfer stages, which is what makes their numbers comparable to the detection stage's. Metric implementations are frozen in a single `eval/metrics.py`; all intervals are 95% percentile bootstrap at $2{,}000$ resamples, stratified by class for binary metrics, and intervals resting on fewer than $2{,}000$ usable resamples are marked as such. The 54 fractal features are declared in advance in `feature_store/p3_feature_manifest.json`; the reliability gate's output, including its empty whitelist and twenty rejected entries, is in `feature_store/p2_reliability_report.json`. The training pipeline (class weighting, prefit sigmoid calibration on the validation fold, threshold derivation from the validation ROC) is implemented once and reused without modification by every downstream script. The one arm not in the original grid — `all_minus_fractal` scored against the edit battery and RAID — was retrained through that same pipeline and confirmed to reproduce the frozen in-distribution AUROC to five decimals before any number from it was used. **One artifact we audited and could not clear**: the 85-feature model's operating-threshold file has four consumers and no writer in the repository, and re-deriving the 1% and 5% FPR thresholds from the frozen model across $\{$calibrated probability, raw probability, raw margin$\} \times \{$validation, test$\}$ reproduces *none* of the stored values, with a gap that is not a rounding miss (6×10^{-4} on the 1% threshold t_1, 1.7×10^{-2} on the 5% threshold t_5). The full audit, including the 5% threshold's staleness and the 1% threshold's internal inconsistency, is in Appendix F; the consequence for this paper is bounded and marked where it bites.

REFERENCES

Ibrahim Alabdulmohsin and Andreas Peter Steiner. A tale of two structures: Do LLMs capture the fractal complexity of language? In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, volume 267 of *Proceedings of Machine Learning Research*, pp. 964–1025. PMLR, 2025. arXiv:2502.14924.

Ibrahim Alabdulmohsin, Vinh Q. Tran, and Mostafa Dehghani. Fractal patterns may illuminate the success of next-token prediction. *Advances in Neural Information Processing Systems*, 37: 112864–112888, 2024. arXiv:2402.01825.

Eduardo G. Altmann, Giampaolo Cristadoro, and Mirko Degli Esposti. On the origin of long-range correlations in texts. *Proceedings of the National Academy of Sciences*, 109(29):11582–11587, 2012.

Guangsheng Bao, Yanbin Zhao, Zhiyang Teng, Linyi Yang, and Yue Zhang. Fast-DetectGPT: Efficient zero-shot detection of machine-generated text via conditional probability curvature. In *International Conference on Learning Representations (ICLR)*, 2024.

Mary J. Cannon, Donald B. Percival, David C. Caccia, Gary M. Raymond, and James B. Bassingthwaighte. Evaluating scaled windowed variance methods for estimating the Hurst coefficient of time series. *Physica A: Statistical Mechanics and its Applications*, 241(3):606–626, 1997.

Yize Cheng, Vinu Sankar Sadasivan, Mehrdad Saberi, Shoumik Saha, and Soheil Feizi. Adversarial paraphrasing: A universal attack for humanizing AI-generated text. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025. arXiv:2506.07001.

Xin Du and Kumiko Tanaka-Ishii. Correlation dimension of auto-regressive large language models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025. arXiv:2510.21258.

Liam Dugan, Alyssa Hwang, Filip Trhlík, Josh Magnus Ludan, Andrew Zhu, Hainiu Xu, Daphne Ippolito, and Chris Callison-Burch. RAID: A shared benchmark for robust evaluation of machine-generated text detectors. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*, 2024. arXiv:2405.07940.

Werner Ebeling and Thorsten Pöschel. Entropy and long-range correlations in literary English. *Europhysics Letters*, 26(4):241–246, 1993.

Jules L. Ellis. The Spearman–Brown formula and reliabilities of random test forms. *arXiv preprint arXiv:2308.13811*, 2023. arXiv:2308.13811.

Sebastian Gehrmann, Hendrik Strobelt, and Alexander M. Rush. GLTR: Statistical detection and visualization of generated text. In *Proceedings of ACL 2019, System Demonstrations*, pp. 111–116, 2019.

Dariusz Grech and Grzegorz Pamuła. On the multifractal effects generated by monofractal signals. *Physica A: Statistical Mechanics and its Applications*, 2013. arXiv:1307.2014.

Yikai Guo, Bin Wang, Xilai Fan, Wenjun Ke, and Haoran Luo. On the salience of low-probability tokens for AI-generated text detection: A multiscale uncertainty perspective. In *International Conference on Machine Learning (ICML)*, 2026. arXiv:2606.02158.

Abhimanyu Hans, Avi Schwarzschild, Valeriia Cherepanova, Hamid Kazemi, Aniruddha Saha, Micah Goldblum, Jonas Geiping, and Tom Goldstein. Spotting LLMs with binoculars: Zero-shot detection of machine-generated text. In *International Conference on Machine Learning (ICML)*, 2024. arXiv:2401.12070.

Ari Holtzman, Jan Buys, Li Du, Maxwell Forbes, and Yejin Choi. The curious case of neural text degeneration. In *International Conference on Learning Representations (ICLR)*, 2020.

Kalpesh Krishna, Yixiao Song, Marzena Karpinska, John Wieting, and Mohit Iyyer. Paraphrasing evades detectors of AI-generated text, but retrieval is an effective defense. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. arXiv:2303.13408.

Lucio La Cava and Andrea Tagarelli. Luminol-AIDetect: Fast zero-shot machine-generated text detection based on perplexity under text shuffling. *arXiv preprint arXiv:2604.25860*, 2026.

Zhichen Liu, Kaitong Qin, Linhan He, and Yang Xu. WaveDetect: Robust framework for machine-generated text detection via wavelet transform. *arXiv preprint arXiv:2606.23336*, 2026.

Ning Lu, Shengcai Liu, Rui He, Qi Wang, Yew-Soon Ong, and Ke Tang. Large language models can be guided to evade AI-generated text detection. *Transactions on Machine Learning Research*, 2024. arXiv:2305.10847.

Sina Mansouri, Mohit Marvania, and Abolfazl Safikhani. Stability-aware feature design for robust watermark detection in machine-generated text. In *International Conference on Machine Learning (ICML)*, 2026. arXiv:2608.18102.

Clara Meister and Ryan Cotterell. Language model evaluation beyond perplexity. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics*, pp. 5328–5339, 2021.

Ginno Millán, Román Osorio-Comparán, and Gastón Lefranc. Preliminaries on the accurate estimation of the Hurst exponent using time series. *arXiv preprint arXiv:2103.02091*, 2021. arXiv:2103.02091.

Eric Mitchell, Yoonho Lee, Alexander Khazatsky, Christopher D. Manning, and Chelsea Finn. DetectGPT: Zero-shot machine-generated text detection using probability curvature. In *International Conference on Machine Learning (ICML)*, 2023.

Thomas Pronk, Dylan Molenaar, Reinout W. Wiers, and Jaap M. J. Murre. Methods to split cognitive task data for estimating split-half reliability: A comprehensive review and systematic assessment. *Psychonomic Bulletin & Review*, 29:44–54, 2022. doi:10.3758/s13423-021-01948-3.

Vinu Sankar Sadasivan, Aounon Kumar, Sriram Balasubramanian, Wenxiao Wang, and Soheil Feizi. Can AI-generated text be reliably detected? Stress testing AI text detectors under various attacks. *Transactions on Machine Learning Research*, 2025. arXiv:2303.11156.

Mihir Sahasrabudhe. Reversal invariance in autoregressive language models. *arXiv preprint arXiv:2511.00341*, 2025.

Shuntaro Takahashi and Kumiko Tanaka-Ishii. Evaluating computational language models with scaling properties of natural language. *Computational Linguistics*, 45(3):481–513, 2019.

Lekkala Sai Teja, Annepaka Yadagiri, Sangam Sai Anish, Siva Gopala Krishna Nuthakki, and Partha Pakray. Modeling the attack: Detecting AI-generated text by quantifying adversarial perturbations. *arXiv preprint arXiv:2510.02319*, 2025.

Brian Tufts, Xuandong Zhao, and Lei Li. A practical examination of AI-generated text detectors for large language models. *arXiv preprint arXiv:2412.05139*, 2024.

Eduard Tulchinskii, Kristian Kuznetsov, Laida Kushnareva, Daniil Cherniavskii, Serguei Barannikov, Irina Piontkovskaya, Sergey Nikolenko, and Evgeny Burnaev. Intrinsic dimension estimation for robust detection of AI-generated texts. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. arXiv:2306.04723.

Dongjun Wei, Minjia Mao, Xiao Fang, and Michael Chau. Short-PHD: Detecting short LLM-generated text with topological data analysis after off-topic content insertion. *arXiv preprint arXiv:2504.02873*, 2025.

Rafał Weron. Estimating long-range dependence: finite sample properties and confidence intervals. *Physica A: Statistical Mechanics and its Applications*, 312(1–2):285–299, 2002. arXiv:cond-mat/0103510.

Alva West, Yixuan Weng, Minjun Zhu, Luodan Zhang, Zhen Lin, Guangsheng Bao, and Yue Zhang. AI-generated text is non-stationary: Detection via temporal tomography. *arXiv preprint arXiv:2508.01754*, 2025.

Zhongxin Yang, Chun Bao, Yuanwei Bin, Xiang I. A. Yang, and Shiyi Chen. Turbulence-like 5/3 spectral scaling in contextual representations of language as a complex system. *arXiv preprint arXiv:2604.05536*, 2026.

A CORPUS-LEVEL GAP: SUPPORTING CHECKS

Table 2: Corpus-level pooled exponents after both corrections; 95% intervals, 10-fold outer bootstrap re-pooling raw trajectories. Human $n = 1{,}000$ per domain, machine $n \approx 21{,}000$–$22{,}700$ per cell, all scored under Gemma-2B. No interval overlaps its domain's human interval on either exponent for either generator: $20/20$. Source: `data/x02_confidence_intervals.json`.

	Hölder S			**Hurst H**		
Domain	Human	`gemma_2b`	`mistral_7b`	Human	`gemma_2b`	`mistral_7b`
`big_patent`	[.4135,.4283]	[.4762,.4806]	[.4929,.5012]	[.6365,.6415]	[.6891,.6911]	[.7123,.7141]
`billsum`	[.3992,.4268]	[.4541,.4638]	[.4478,.4538]	[.6315,.6368]	[.6674,.6693]	[.6859,.6879]
`newsroom`	[.3539,.3794]	[.4627,.4668]	[.4888,.4917]	[.6021,.6122]	[.6613,.6635]	[.6893,.6918]
`scientific_papers`	[.3896,.4046]	[.4659,.4719]	[.4965,.5037]	[.6068,.6118]	[.6864,.6889]	[.7131,.7156]
`wikipedia`	[.3944,.4301]	[.4428,.4505]	[.4657,.4705]	[.6208,.6279]	[.6550,.6569]	[.6754,.6773]

Two further checks on §3.1's $20/20$ separation, plus a replication failure on the source publication's separate *spread* claim.

The duplicate-count finding itself: GAGLE's split label had been treated as part of an article's identity, so the same article recurred as several "different" human documents. Every one of the $4{,}474$ cross-split copies checked is byte-identical, and collapsing all duplicate groups takes the human population from $\approx 11{,}700$ rows to $5{,}000$ articles, a $2.34\times$ over-count. The $4{,}474$ is the byte-identity verification count, not the number of rows removed; the reduction itself is the $\approx 6{,}700$ excess copies.

1. **The population correction behaves as theory predicts.** If the pre-correction human intervals were narrow *because of* duplication, removing duplicates should widen them by $\sqrt{N_{\text{pre}}/N_{\text{post}}}$. Measured widening ratios run 1.27–1.61 with mean 1.455 against a predicted 1.47–1.57 (mean 1.529), and $20/20$ cells widen. The measured mean sits 5% below prediction, the direction expected when the duplicates were byte-identical rather than independent draws.
2. **The direction is not the one a "models lack long-range planning" story predicts.** Machine text has *higher* H than human text in every cell. Under the estimator's own semantics — $H \to 1$ is strong persistence, in the limit verbatim repetition — higher H is consistent with the repetition and degeneration behaviours documented for autoregressive decoding (Holtzman et al., 2020; Du & Tanaka-Ishii, 2025), not with richer discourse planning. We flag this as an interpretation of the estimator's semantics, not as a measured mechanism; §8 states what would be needed to establish it.

A replication failure worth recording. The source publication additionally reports that human writing has a *narrower* spread of S than model output. On this corpus the direction reverses, and it survives a length control: the variance ratio $\text{Var}_{\text{LLM}}/\text{Var}_{\text{hum}}$ of S exceeds 1 significantly in 8 of 10 (domain, scorer) cells (largest 1.457, `scientific_papers`/Mistral), with the two dissenting cells (`billsum`) null rather than reversed; residualising on $\log$ document length moves every ratio by at most 0.016, and the direction holds in $5/5$ length quintiles for both scorers. Bootstrap and Brown–Forsythe agree on direction in every cell. *The same is not true of* H: raw H spread ratios are all significant at 1.19–1.89, but after length residualisation only six remain significant and three reverse, two of them significantly (`wikipedia`/Mistral $1.243 \to 0.813$; `big_patent`/Mistral $1.242 \to 0.883$). An H spread difference should not be reported as a structural property without the length adjustment.

B FULL SPLIT-HALF RELIABILITY TABLE

All twenty candidate quantities reached the gate and none passed. The twelve reported in the source write-up are reproduced below; the remaining eight, including the fit-quality diagnostics S_R^2 and H_R^2, are in `feature_store/p2_reliability_report.json`. R^2 rows are fit-quality diagnostics rather than features, and their low reliability is expected.

Table 3: Split-half reliability, all reported rows. Bar $\rho \geq 0.60$. Source: `reports/2026-08-23_P2_split_half_reliability.md`.

Quantity	n	r (half)	ρ (S–B)	Verdict
`gemma_H`	221,904	0.405	0.576	drop (near miss)
`mistral_H`	223,834	0.376	0.547	drop
`gemma_H_window_mean`	221,904	0.275	0.431	drop
`mistral_H_window_mean`	223,834	0.247	0.396	drop
`gemma_H_window_drift`	221,904	0.192	0.322	drop
`mistral_H_window_drift`	223,831	0.167	0.286	drop
`gemma_H_window_var`	221,904	0.162	0.279	drop
`mistral_H_window_var`	223,834	0.158	0.273	drop
`gemma_S`	196,971	0.032	0.063	drop
`mistral_S`	195,552	0.025	0.050	drop
`gemma_S_window_mean`	122,329	0.008	0.017	drop
`mistral_S_window_var`	116,669	0.002	0.003	drop

Method detail. Halves are 200 tokens, so they use the reduced scale sweep $\{8, 16, 32, 48, 64, 96\}$ already validated for length-200 series; the full sweep reaches $\tau = 320$, which is meaningless on 200 points. Windowed quantities use a reduced $100/50$ window because the production $200/100$ window yields exactly one window on a 200-token half, making drift undefined and variance degenerate — they therefore measure the stability-profile *concept*, not the exact production column. Per-column masking with no row dropping, so each row reports its own n. The reduced n on the $\widehat{S}$ rows (196,971 of 223,834, $\approx$ 120,000 on the windowed variants) is itself informative: on those documents at least one half produced a degenerate Hölder fit.

B.1 Estimator bias against ground truth, full ladder

§3.2 summarises this as a document-support bias of -0.26 to -0.48; the full two-point ladder against synthetic fractional Brownian motion of known exponent, run before the corpus was built:

Table 4: $\widehat{S}$ bias against known synthetic ground truth, by support. Source: `reports/2026-07-26_S_estimator_fix.md`.

Support	$\widehat{S}$ bias vs. truth	fit R^2
One 400-token document, $n_{\text{boot}} = 1000$–5000	-0.26 to -0.48	0.08–0.38
$\approx$80,000 points (200×400 concatenated), $n_{\text{boot}} = 2000$	-0.004 to -0.065	0.91–0.995

Raising the number of bootstrap draws at fixed support does not close that gap: more draws do not fix a short window. In the pooling mode the corpus-level numbers actually use, bias grows with the true exponent — -0.072, -0.160, -0.231 and -0.310 at $H = 0.30$, 0.50, 0.65 and 0.80.

B.2 Reliability under alternative split constructions

The full run behind §3.2's split-construction test: eight constructions, all four headline quantities, both the pooled and the stratum-demeaned within-population coefficient (§B.3 defines ρ_{within}).

The headline robustness test. The plan's own pre-registered bar asks whether `gemma_S` stays inside $[0.03, 0.12]$ across every valid construction. It does not: `contiguous` falls to 0.0159. That is the *expected* signature of the drift confound the interleaved design was built to avoid — `gemma_H` collapses in the same construction, $0.576 \rightarrow 0.261$, consistent with genuine within-document stylistic drift entering a contiguous split and not the other constructions — so it is read as confirming, not undermining, the case for interleaving. The five remaining valid constructions (interleave, four block sizes, the permutation median and the matched full sweep) span 0.0396–0.0626, comfortably inside the band.

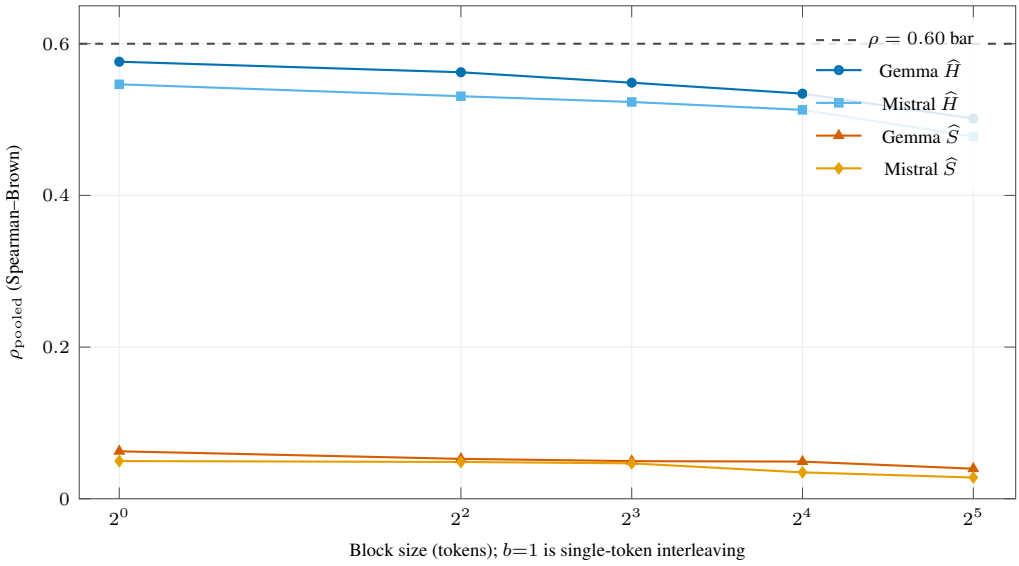


Figure 2: Reliability against block size, $b = 1$ is single-token interleaving. The decline is smooth and monotonic in the direction that argues *against* the sequence-geometry objection: coarser blocks, which decimate the sequence less, are systematically *less* reliable, not more. Source: `review_results/data/t0_t1_MERGED_table.json`.

Table 5: Split-half reliability (ρ_{pooled}) under all eight tested constructions. `block_`b assigns contiguous blocks of b tokens alternately to the two halves, so `interleave` is the $b = 1$ case and `contiguous` the single-block limit. `block_bootstrap` is measured but structurally invalid here: its two halves share $\approx 47\%$ of token positions by construction (block length ≥ 96 on a ≈ 400-token document), so its inflated correlation reflects token overlap, not document-level signal, and it is excluded from the headline claim. `permutation` is the median of 20 random-membership replicates (order preserved within each half), [p5,p95] in brackets. Source: `review_results/data/t0_t1_MERGED_table.json`, `t1_permutation_stats.json`.

Construction	gemma_H	gemma_S	mistral_H	mistral_S
interleave (headline)	0.5763	0.0626	0.5465	0.0497
block_4	0.5624	0.0526	0.5308	0.0485
block_8	0.5486	0.0496	0.5232	0.0467
block_16	0.5342	0.0490	0.5128	0.0348
block_32	0.5013	0.0396	0.4777	0.0280
contiguous (pre-registered as confounded)	0.2609	0.0159	0.2678	0.0040
permutation, 20 reps (median [$p5$, $p95$])	0.5443 [.5424, .5462]	0.0530 [.0484, .0571]	0.5152 [.5129, .5182]	0.0452 [.0386, .0498]
matched_sweep (full 6-value `WINDOW_SCALE`)	0.5763	0.0626	0.5465	0.0497
`block_bootstrap` (excluded, see caption)	0.7779†	0.2047†	0.7769†	0.1912†

An order-scrambling construction, kept as a diagnostic. Among the constructions swept for the permutation check is one built from an *unsorted* random index set for each half, which scrambles a half's internal token order along with its membership. Because both estimators are functionals of ordered structure, this construction collapses ρ to ≈ 0.003 for every quantity, in sharp contrast to every order-preserving construction tested. We keep it here as a direct demonstration that these quantities are genuinely order-dependent — destroying order destroys the signal, which a magnitude or length statistic computed on the same tokens would not do.

B.3 Variance decomposition (Route 1 and Route 2)

Route 1 demeans both halves by their own (class, domain, temp) stratum before correlating — the number quoted in §2 as ρ_{within}. Route 2 is a closed-form, one-way variance-components decomposition (method-of-moments, not an iterative mixed model, chosen for zero convergence risk at $n \approx 223{,}834$) that additionally reports the population share and a cluster-robust standard error, clustered by seed article.

Table 6: Variance decomposition. ICC is the fraction of single-half variance explained by (class, domain, temp) stratum membership, i.e. an estimate of $\mathrm{Var}(\mu)/(\mathrm{Var}(\mu)+\mathrm{Var}(\delta)+\mathrm{Var}(\eta))$. Source: `review_results/data/t0_variance_components.json`.

Quantity	ρ_{pooled}	ρ_{within}	ICC	$\mathrm{Var}(\delta)$	cluster-robust SE
`gemma_H`	0.576	0.552	3.9%	0.00217	0.0023
`mistral_H`	0.547	0.520	3.9%	0.00190	0.0024
`gemma_S`	0.063	0.060	0.14%	0.01049	0.0024
`mistral_S`	0.050	0.046	0.18%	0.00870	0.0023

Reading the two exponents side by side: for $\widehat{H}$, population structure (class, domain, temperature together) explains only $\approx 4\%$ of single-half variance, so the $0.576 \to 0.552$ drop is real but modest — most of what §3.2 measures for $\widehat{H}$ was already within-population. For $\widehat{S}$, population structure explains under 0.2% of variance: $\widehat{S}$'s poor pooled reliability is essentially all within-document estimation noise, ruling out "pooling across populations inflates apparent unreliability" as an explanation for $\widehat{S}$ either way. *Note*: GAGLE has no generator column distinct from `Label` (Human/Gemma-2B/Mistral-7B already is the class–generator join), so the demeaning set is (Label, domain, temp) throughout — the same information content as (class, generator, domain, temp), without a redundant fourth column. Prompt type is deliberately excluded from the demeaning set throughout this appendix, because §4.2 treats what is predictable from prompt type as part of δ; demeaning by it here would remove the very quantity that section measures.

B.4 Reliability as a function of support, and $N^\star$

The construction behind §3.2's $N^\star$ claim. An estimator noise floor $\mathrm{Var}(\eta_N)$ is calibrated against synthetic fractional Gaussian noise (Davies–Harte generator) at $H \in \{0.60, 0.65, 0.70, 0.72\}$, averaged to match GAGLE's own observed $\widehat{H}$ band; $\mathrm{Var}(\delta)$ is taken from Table 6; and $\rho_{\text{SB}}(N) = 2r(N)/(1+r(N))$ for $r(N) = \mathrm{Var}(\delta)/(\mathrm{Var}(\delta)+\mathrm{Var}(\eta_N))$ is composed at each grid point $N \in \{100, 200, 400, 800, 1600, 3200, 6400, 12800\}$. $N^\star$ is the first grid point at which $\rho_{\text{SB}}(N)$ crosses 0.60; the true crossing may lie between grid points.

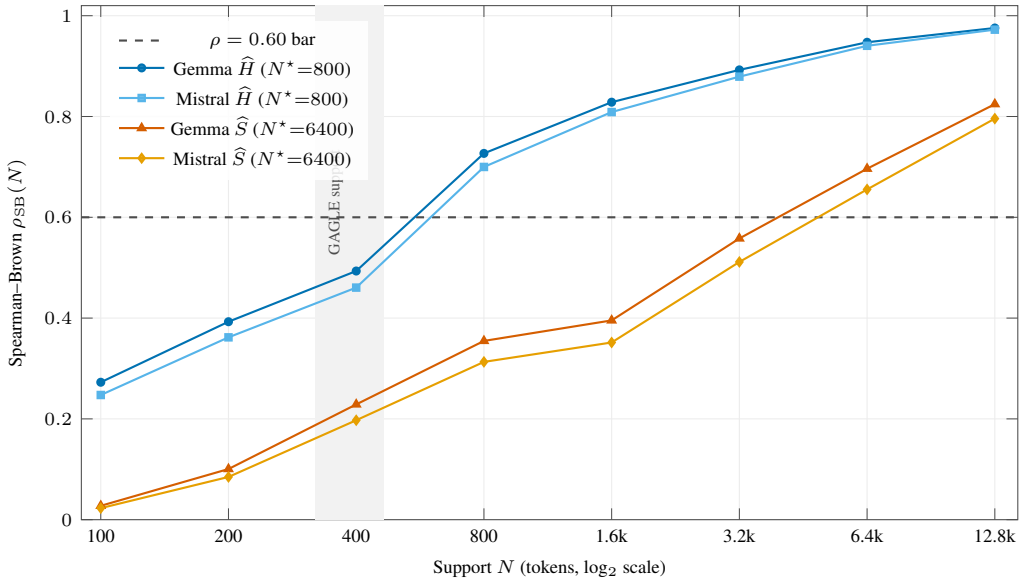


Figure 3: Reliability against support, calibrated fBm noise floor. Shaded band: GAGLE's own 320–464-token usable-support range. $\widehat{H}$ (blue, sky) crosses $\rho = 0.60$ near N=800; $\widehat{S}$ (vermillion, orange) does not cross until N=6,400. Source: `review_results/data/t2_n_star_composition.json`.

Table 7: $N^\star$: first grid point where the calibrated $\rho_{\text{SB}}(N)$ crosses 0.60. Full curve (all eight grid points) in `review_results/data/t2_n_star_composition.json`.

Quantity	$\text{Var}(\delta)$	ρ_{SB} at N=400	$N^\star$
`gemma_H`	0.00217	0.493	**800**
`mistral_H`	0.00190	0.460	**800**
`gemma_S`	0.01049	0.229	**6,400**
`mistral_S`	0.00870	0.197	**6,400**

What the real-text bracket does and does not establish. A companion pass measures within-document block variance of $\widehat{H}/\widehat{S}$ across disjoint N-token blocks on 5,000 full-length human documents (uncapped up to the corpus's own 2,047-token limit). At the matched N values this real-text variance runs at or below the calibrated fBm floor (e.g. at $N = 400$: $\widehat{S}$ variance 0.0603 real-text against 0.0707 fBm; $\widehat{H}$ variance 0.0046 against 0.0045, materially the same). An earlier draft of this analysis read that gap as an upper bound on $\text{Var}(\eta_N)$; it is not — blocks drawn from one document share that document's own realisation, which suppresses between-block variance relative to the independent draws the fBm floor uses, an effect distinct from and opposed to the nonstationarity a bound argument would need. What the bracket *does* establish, via a calibration check (median full-document $\widehat{H} = 0.629$ across all 5,000 documents, squarely inside the 0.60–0.72 calibration band), is that the fBm floor is not an underestimate of real-text estimator scatter *for* $\widehat{H}$. **The check does not extend to $\widehat{S}$, and there the floor is measurably optimistic**: the composed $\rho_{\text{SB}}(400)$ of 0.229/0.197 (Table 7) sits 3.7–4.0× above the 0.063/0.050 measured at the same support (Table 3), so it understates $\text{Var}(\eta_{400})$ for $\widehat{S}$. $N^\star$ is therefore a point estimate under a validated floor for $\widehat{H}$, and a *lower* bound for $\widehat{S}$: correcting the floor moves $\widehat{S}$'s crossing further out, never nearer.

B.5 Estimator invariance, part 1: detrended fluctuation analysis and local Whittle

This appendix and the persistent-homology pilot of Appendix L ("Estimator invariance, part 2") ask the same question with three structurally different tools: is the short-support reliability ceiling a property of *what* is being measured, or an artifact of *how* one particular estimator measures it? DFA and the local-Whittle spectral estimator are each designed, for different reasons, to be more robust than the plain rescaled-range (R/S) statistic used as this paper's headline to the kind of nonstationarity a short, topically drifting document introduces. DFA fits and subtracts a local polynomial trend from the surprisal walk before measuring its fluctuation at each scale, so slow within-document drift does not get folded into the exponent; the local-Whittle estimator instead works in the frequency domain, fitting the low-frequency slope of the sequence's power spectrum, which is less sensitive to a handful of extreme surprisal spikes than a windowed range statistic is. If either recovers substantially higher reliability than R/S at the same support, the ceiling would be an R/S artifact rather than a property of the estimand; if not, it generalises.

Refitting DFA (order-1, same scale sweep) and a local-Whittle spectral estimator on the identical interleaved halves used throughout this appendix.

Table 8: Reliability by estimator. `passes` marks $\rho \geq 0.60$. Local-Whittle saturation (its bounded optimiser hitting $[-0.49, 1.49]$) affects 0.0001 of pairs for both scorers and moves reliability by $<$ 0.001 when excluded — not the source of its clearance. Source: `review_results/data/t4_alt_estimators_reliability.json`.

Quantity	n	ρ_{pooled}	ρ_{within}	Passes 0.60?
`gemma_RS_H` (headline)	221,904	0.576	0.552	no
`gemma_DFA_H`	221,904	0.629	0.601	**yes (by 0.029)**
`gemma_Whittle_H`	221,904	0.677	0.647	**yes**
`mistral_RS_H` (headline)	223,834	0.547	0.520	no
`mistral_DFA_H`	223,834	0.599	0.566	no (by 0.0006)
`mistral_Whittle_H`	223,834	0.642	0.605	**yes**

Table 9: Cross-estimator correlation, document level (mean of halves). Thresholds fixed before this table was produced: > 0.85 reads as "the same quantity, more precisely measured"; < 0.60 reads as "a different quantity". Source: same file as Table 8.

Pair	Gemma	Mistral
R/S $\sim$ DFA	0.869	0.864
R/S $\sim$ Whittle	0.759	0.751
DFA $\sim$ Whittle	0.808	0.803

R/S and DFA clear the "same quantity" threshold (0.86–0.87): DFA is a less-noisy estimator of the same Hurst exponent, and its clearance of 0.60 is real but marginal and scorer-dependent, which is why §3.2 reports it as "at the margin" rather than as unqualified recoverability. R/S and Whittle correlate more weakly (0.75–0.76), so Whittle's more comfortable clearance (0.677/0.642) is partly a different quantity, not purely a cleaner measurement of the same one, and is reported here alongside this correlation rather than as a standalone headline. Neither estimator has a literature-standard analogue for $\widehat{S}$'s displacement-probability construction; $\widehat{S}$'s reliability is unaffected by this appendix and stands exactly where §3.2 puts it.

Part 2 of this argument, on a representation with no scale-exponent machinery at all, is in Appendix L ("Estimator invariance, part 2"): a persistent-homology pilot on token-embedding trajectories hits the same short-support wall by an entirely different route — evidence for the mechanism itself, not for an artifact of any one estimator's implementation.

C Extended Detection Results

Harness controls. The label-shuffle control runs the full harness on permuted training labels and lands systematically *below* chance — mean AUROC 0.3693 (sd 0.0427) on the random fold, $0/10$ permutations above 0.5, and all eight folds pass a one-sided criterion — because human rows occupy a sparse extreme region of the feature space that a class-weighted booster on random labels assigns the global prior. Injecting leakage raises it monotonically and past the decision boundary at every non-zero level tested (0%: 0.4166; 2%: 0.5080; 5%: 0.5534; 20%: 0.8019), so the control detects what it exists to detect, and its sensitivity floor sits below a 2% contamination rather than above 5%. This catches a class of failure — an index misalignment, or a metric computed against the wrong vector — that no confound configuration could, because all five would be wrong together.

Multiplicity, and the learner check. Across all forty paired tests (five configurations $\times$ eight folds) DeLong and the paired bootstrap agree in 40 of 40, and Holm–Bonferroni leaves *three* survivors, all on the `billsum` fold and cutting both ways — two where fractal features are significantly worse ($p = 9.4\times10^{-9}$, $p = 3.2\times10^{-5}$) and one where they are significantly better ($p = 4.6\times10^{-5}$), across three configurations. A cross-fold "fractal is significantly harmful" framing does not survive correction; what survives is narrower and more interesting — the effect is statistically real only where the domain shift is largest, and its sign there depends on how missingness is handled. The sole cell favouring the full model, `lodo/big_patent`, does so by 0.0001 AUROC, inside that cell's own bootstrap width. The result is not an artifact of the learner: a shallow multilayer perceptron (one hidden layer, 64 units, imputed and scaled, class-balanced) trails the tree ensemble on all eight folds and collapses to TPR@1%FPR $= 0.0000$ on two of them, where the tree ensemble never does, and the fractal family does not rescue it.

The substitution test, per fold. Isolating a nine-feature log-perplexity group and running `fractal+logppl` (63 features) against `stat_only` (17) on population-matched folds, the fractal arm loses on all eight of eight folds on both AUROC and TPR@1%FPR: $0.9684/0.6460$ against $0.9795/0.7289$ in-distribution, and $0.8191/0.2050$ against $0.8735/0.3171$ on `billsum` LODO. Swapping fractal features *in* for entropy and rank statistics, at comparable combination shape, makes every fold worse.

Remaining arms. Three arms omitted from Table 10 for space: `fractal+style` (68 features) reaches AUROC 0.9917 / TPR@1%FPR 0.8171, within noise of `style_only` on AUROC; `all_`–

Table 10: Feature-family ablation, in-distribution (random split, `full`), $n_{\text{test}} = 43{,}413$, human prevalence 2.23%, 2,000-fold stratified bootstrap. Remaining arms and all eight folds: Appendix C.

Arm	#feat	AUROC	TPR@1%FPR	Reading
`fractal_only`	54	0.6629 [.6458, .6782]	0.0641 [.0429, .0914]	real signal, far below either baseline
`stat_only`	17	0.9795	0.7209	incumbent: perplexity, entropy, ranks
`style_only`	14	0.9919	0.7964	incumbent: discourse and surface form
`fractal+stat`	71	0.9785	0.6836	*below* `stat_only` on both
`all`	85	0.9984 [.9976, .9990]	0.9587 [.9489, .9738]	full model
`all_minus_fractal`	31	0.9984	0.9612 [.9360, .9781]	**all 54 fractal features deleted**

`minus_length` (83) reaches 0.9974/0.9436, so removing length costs 0.0010 AUROC and 0.0151 TPR@1%FPR; and `length_only` (2) reaches 0.8635/0.0477, a partial detector of *human* text and the priced form of the corpus's dominant confound. Four further diagnostic arms, each of which declared a failure condition in advance, on the same random split and `full` configuration: `provenance_only` reaches AUROC 0.5655 / TPR@1%FPR 0.1196 on the six window-length provenance columns, which prices that exclusion (the eleven exit-gate flags are a separate, disjoint set); `shortwindow_only` (2) reaches 0.5290/0.0580, weakly predictive and demonstrably unused; `missingness_only` (85) reaches 0.5169/0.0305, the missingness pattern alone; and `label_shuffle` lands at 0.3693 with $0/10$ permutations above chance.

Table 11: `all` versus `all_minus_fractal` on all eight frozen group-disjoint folds, `full` configuration: one random in-distribution split, five leave-one-domain-out (LODO) and two leave-one-generator-out (LOGO). Each fold is scored on the frozen test split of its held-out stratum, which is why the LOGO n is a fraction of that generator's corpus count rather than all of it. AUROC at six decimals to make the ties legible. `all_minus_fractal` ties or wins on 7 of 8. Source: `data/p3_t1_results.json`; `reports/2026-08-30_fractalscope_verification_checks.md` §2.

Scheme	Held-out fold	n_{test}	`all` (85)	`a-m-f` (31)	Δ	Winner
random	—	43,413	0.998397	0.998397	$+1.5\times10^{-7}$	tie
lodo	`big_patent`	45,175	0.998571	0.998468	-0.000103	`all`
lodo	`billsum`	45,146	0.937608	0.941725	$+0.004117$	no-fractal
lodo	`newsroom`	44,718	0.992131	0.992077	-0.000054	tie
lodo	`scientific_papers`	43,749	0.994487	0.995077	$+0.000590$	no-fractal
lodo	`wikipedia`	45,046	0.987009	0.987990	$+0.000981$	no-fractal
logo	`gemma_2b`	22,747	0.987404	0.988510	$+0.001106$	no-fractal
logo	`mistral_7b`	21,635	0.986638	0.987624	$+0.000986$	no-fractal

Table 12: AUROC / TPR@1%FPR, `full` configuration, gradient-boosted tree, all arms on the seven shifted folds. Source: `results/p3_tables.md`, LODO and LOGO config 1.

Scheme	Fold	`fractal` (54)	`stat` (17)	`style` (14)	`all` (85)	`a-m-f` (31)	`length` (2)
lodo	`big_patent`	.6274/.0278	.9626/.6034	.9907/.7617	.9986/.9632	.9985/.9539	.9734/.2636
lodo	`billsum`	.5278/.0311	.8732/.3235	.8797/.0612	.9376/.4931	.9417/.5031	.7234/.0177
lodo	`newsroom`	.6624/.0406	.9858/.8143	.9385/.3570	.9921/.8921	.9921/.8847	.7566/.0077
lodo	`scientific_papers`	.6968/.0913	.9810/.7705	.9851/.6336	.9945/.8769	.9951/.8918	.9529/.0819
lodo	`wikipedia`	.5984/.0464	.9667/.7245	.9561/.4518	.9870/.8178	.9880/.8345	.8132/.0359
logo	`gemma_2b`	.6163/.0531	.9004/.4171	.9786/.6112	.9874/.7954	.9885/.8079	.8506/.0333
logo	`mistral_7b`	.6567/.0746	.9089/.4114	.9810/.7065	.9866/.7419	.9876/.7555	.8549/.0555

Two patterns in Table 12 are worth naming. First, `fractal_only` is at or near chance on the hardest fold (`billsum`, AUROC 0.5278) and never exceeds 0.6968 anywhere. Second, the incumbent families trade places by domain — `style_only` leads on `big_patent` and `scientific_papers`, `stat_only` on `newsroom` — the signature of a detector whose components are domain-sensitive in different directions, and part of why the fused model holds up better than any single family.

The `billsum` LODO error analysis. The hardest fold shows the failure mode the whole 85-feature representation has, independent of the fractal family. AUROC falls only to 0.9376 while TPR@1%FPR falls to 0.4931 — a tail-ranking failure, not a calibration artifact: bulk ranking is preserved and the extreme tail is not. An error analysis on this fold (run on a refit whose headline cell was 0.9476/0.5630, before a later feature-tokenisation correction moved it to the values quoted above; the mechanism, not the level, is what carries forward) finds the driver is prompt type by a wide margin. False-negative rate runs 0.677 for chain-of-thought, 0.676 for excerpt continuation and 0.608 for pretrained continuation, against 0.259–0.350 for keyword- and summary-constrained generation — a 1.7–2.6× gap. Freely generated text on this domain's legal-summary register reads closer to human. The largest distributional shifts into this fold are style features (`Sent_Len_Std` +0.595 standardised, KS 0.255; `Bigram_Repetition_Rate` +0.263) — the family carrying 63.46% of the model's gain.

Deployment-operating-point diagnostics. Beyond the ranking metrics above, three questions bear on deployment: how the detector behaves at very low false-positive budgets, whether its threshold calibration holds under resampling, and how precision changes with the assumed prevalence of LLM-generated text in the traffic being screened. On the `full` configuration random split ($n = 43{,}413$), partial AUROC restricted to FPR$\leq 1\%$ is 0.9595 and to FPR$\leq 5\%$ is 0.9857; TPR at a fixed 5% FPR threshold is 0.9952, and at a much stricter 0.1% FPR threshold falls to 0.6729 (point estimate, 2,000-fold bootstrap). Re-deriving the operating threshold on resampled folds and checking the FPR it actually achieves against its 1% and 5% targets gives 0.93% [0.41%, 1.55%] and 4.95% [3.61%, 6.40%] respectively — both targets covered within CI, so the thresholds in Table 10 are not overfit to one split. Expected calibration error (15 equal-width bins) is 0.00091: the model's predicted probabilities track observed frequencies closely at this operating point.

GAGLE's own corpus is 97.77% LLM-generated (human text is the rare class by construction of the benchmark); a real moderation deployment is typically the opposite shape, with LLM-generated content the minority of traffic. Because precision is prevalence-dependent even when AUROC is not, Table 13 reports PPV at the 1%- and 5%-FPR thresholds across a range of *assumed* LLM prevalence in deployment traffic, not GAGLE's own training prevalence, up to GAGLE's own composition as a reference point.

Table 13: Positive predictive value at fixed FPR thresholds, by assumed LLM-content prevalence in deployment traffic (distinct from GAGLE's own training-corpus composition, bottom row, included as a reference point). Each cell is $\mathrm{PPV} = \mathrm{TPR}\,\pi / \big(\mathrm{TPR}\,\pi + \mathrm{FPR}(1-\pi)\big)$ at the achieved FPRs (0.93% and 4.95%), with π the assumed prevalence. `full` configuration, random split.

Assumed LLM share of traffic	PPV @ 1%FPR	PPV @ 5%FPR
1%	0.510	0.169
5%	0.845	0.514
10%	0.920	0.691
25%	0.972	0.870
50%	0.990	0.953
97.77% (GAGLE's own composition)	0.9998	0.999

D Feature Importance Accounting

Split gain and permutation importance are reported together throughout, because they disagree. Table 14 gives the family totals; the per-column table for all 85 inputs, including every zero-contribution column, is in `results/p3_tables.md`. Reporting only the columns that worked is the selective reporting the pre-declared reporting rule exists to prevent, so every declared feature appears there, and none has exactly zero gain.

Per-column detail. Only two fractal columns appear in the top 25 by gain. The highest, `Gemma_H`, ranks 23rd with 0.28% of gain and a permutation drop of $3{\times}10^{-5} \pm 3{\times}10^{-5}$; for comparison the top column overall, `N_Words_Used`, has 39.99% of gain and a drop of $1.28{\times}10^{-2} \pm 2.1{\times}10^{-4}$. The two cross-scorer divergence features — the design's explicit attempt to extract signal from *disagreement* between a 2B and a 7B scorer's estimates on the same document — rank 60th and 68th of 85.

Table 14: Gain versus permutation accounting, `all` arm, random split. Permutation importance is the AUROC drop when a column is shuffled on the test fold, five repeats. Source: `results/p3_tables.md` (D-13).

Family	#feat	total split gain	gain share	summed perm. drop	perm. share
style	14	1,296,092.4	63.46%	2.04×10^{-2}	56.7%
stat	17	633,456.7	31.02%	1.54×10^{-2}	42.8%
fractal	54	112,761.3	5.52%	$\mathbf{1.7 \times 10^{-4}}$	**0.47%**

Distribution within the fractal family. All 54 columns have a permutation drop below 10^{-4}. Thirty-five are zero or negative — shuffling them improves held-out AUROC — and thirty-eight lie within their own standard error of zero. The highest-ranked fractal column by gain is `Gemma_H` at overall rank 23 (0.28% gain, 237 splits, permutation drop $3 \times 10^{-5} \pm 3 \times 10^{-5}$); the lowest is `Mistral_S_window_max` at rank 85. Only two fractal columns appear in the top 25.

Reading the disagreement. Gain counts splits weighted by local objective improvement, which a greedy learner will accumulate on any column with exploitable within-node variation, informative or not. Permutation measures what the fitted model loses on held-out data without the column. For a column that is noise around a population-conditional mean the first is substantial and the second is nil, which is exactly the pattern observed. The same caution applies in the other direction and we apply it symmetrically: `N_Words_Used` holds 39.99% of gain, but removing length from the model costs only 0.0010 AUROC and 0.0151 TPR@1%FPR (§4). Gain share overstates necessity in both cases.

E Perturbation Battery: Full Detector Comparison and Arm Decomposition

Table 15: Recall@1%FPR with and without the fractal family, by operator class. *The class predicts the sign of the contribution in all six cases.* Row counts differ slightly because the 85-feature numbers use the Binoculars-limited 39,773-row intersection and the 31-feature numbers the full 39,810-row edited store. Source: `data/p4_task5_final_comparison.json`.

Class	Operator	n (85)	n (31)	`all` (85)	`a-m-f` (31)	Contribution of fractal family
–	(pre-attack, clean)†	42,444	42,444	0.9559	0.9597	−0.38 pt (tie; see caveat)
II	`paraphrase_light`	5,157	5,162	0.9639	0.9545	+0.94 pt
	`paraphrase_heavy`	5,255	5,265	0.9243	0.9033	+2.10 pt
	`style_transfer`	5,380	5,393	0.9520	0.9425	+0.95 pt
	`summarize_then_expand`	4,293	4,298	0.9846	0.9744	+1.02 pt
I	`sentence_reorder`	9,831	9,835	**0.2670**	**0.2866**	**−1.96** pt
	`typo_noise`	9,857	9,857	**0.5851**	**0.6727**	**−8.76** pt

Why a sign flip is the predicted outcome. Under $\mathrm{Var}(\eta) \gg \mathrm{Var}(\delta)$ a large structured perturbation enters an already noise-dominated estimate, the learned thresholds read the wrong side, and a model that never held the quantity is better off than one holding a corrupted version of it. `sentence_reorder` preserves every token, every word frequency and every lexical statistic while permuting the order in which they arrive; `typo_noise` preserves the content and perturbs the tokenisation. Everything else in the model is a content statistic, and content is exactly what these two operators preserve — which is why they, and only they, invert the family's contribution.

Operator definitions. *Class I (index structure).* `sentence_reorder` permutes contiguous blocks, destroying the ordering the rescaled range integrates over; `typo_noise` performs character-level swaps, drops and slips that re-tokenise the affected words, shifting every downstream index and injecting high-surprisal spikes into the increment sequence. *Class II (realisation).* `paraphrase_light`, `paraphrase_heavy`, `style_transfer` and `summarize_then_expand` rewrite vocabulary and register at fixed coarse-scale ordering.

The two qualifications on §5. An arm decomposition localises the Class I *collapse* in the non-fractal families, `fractal_only` being already at 5.8% pre-attack and barely moving. And the pre-attack row † of Table 15 is a copy-through from an operating-threshold file we could not re-derive, whereas all twelve per-attack recall values (six operators × two arms) reproduce *bitwise* from the stored thresholds; re-deriving them narrows `typo_noise`'s contribution from −8.76 to roughly −7 points without approaching zero.

The operator partition generalises beyond this estimator family: for any representation defined as a functional of some structure in the input, an operator acting on that structure and one acting around it constitute a two-sided test of whether the representation encodes the structure or merely correlates with the population that produced it.

Table 16: Recall@1%FPR by detector and operator, shared population $n = 39{,}773$. ↑ marks recall *above* the detector's own pre-attack baseline. Source: `data/p4_task5_final_comparison.json` via `results/p4_robustness.md`.

	Pre-attack		**Class II: realisation**				**Class I: index**	
Detector	@1%	@5%	para. light	para. heavy	style	summ.→exp.	reorder	typo
`roberta`	0.9989	0.9999	0.9996	0.9989	0.9993	0.9991	0.9959	0.9968
`tabular` (85)	0.9559	0.9906	0.9639	0.9243	0.9520	0.9846	**0.2670**	**0.5851**
`binoculars`	0.8728	0.9231	0.8689	0.7922	0.8554	0.7456	0.7846	0.8110
`fastdetectgpt`	0.5104	0.6482	0.6735↑	0.6344↑	0.6898↑	0.8600↑	0.2430	0.1967
`perplexity_gemma` *(floor)*	0.0935	0.3339	0.0076	0.0076	0.0063	0.0005	0.0183	0.0041

One caveat is stated once rather than per row: a detector whose *pre-attack* recall is already near the floor (`perplexity_gemma`, 0.0935) has no signal for an attack to destroy or spare, and its post-attack numbers describe noise around an uninformative baseline, not robustness.

Arm decomposition, in full. Refitting single-family arms on the detection stage's own training rows (`config=full`, random/random) and scoring them against `sentence_reorder`: `all_minus_fractal` (31 features, stat and style pooled) falls 94.3% → 22.7% recall@1%FPR, a 71.6-point drop; `stat_only` (17) falls 80.6% → 34.5% (46.1 points); `style_only` (14) falls 77.6% → 27.6% (50.0 points); and `fractal_only` (54) was already at 5.8%/17.4% (at the 1% and 5% operating points) pre-attack and barely moves under attack (3.2%/14.2%) — a floor effect, not evidence of robustness. Two qualifications, because the arithmetic invites an overclaim in each direction. **(i)** The drops are *not* additive: at 5% FPR the pooled arm degrades *less* than `style_only` alone (39.0 against 44.0 points), so pooling gives the model redundant signal to fall back on at that operating point rather than simply summing two overlapping weaknesses — which is why we report post-attack absolute recall alongside drop size rather than asserting an additivity pattern the numbers do not support. **(ii)** These arms are refits on the detection stage's own rows, a different population from the 39,773-row shared intersection of Table 16, so they are read for direction and relative magnitude, not merged with that table's levels.

F Operating-Threshold Provenance: What We Could and Could Not Re-Derive

The 85-feature detector's operating-threshold file is read by four scripts and written by none. We attempted to re-derive it rather than restate the caveat.

Q1 — derivation: no convention reproduces the stored values. Loading the frozen `full/random/random/all/lgbm` model and re-deriving the 1% and 5% FPR thresholds across three score conventions (calibrated probability, raw probability, raw margin) crossed with two splits (validation, test) gives six candidates, none of which matches. Calibrated probability is closest in scale, and it is not a near miss: the residual is $\approx 6 \times 10^{-4}$ on t_1 and $\approx 1.7 \times 10^{-2}$ on t_5. Row-membership and quantile-convention scans across all 223,834 rows returned nothing. Version control does not help: the data directory is excluded from the repository, so the file was never tracked and this is not a recoverable deletion. Nor does any surviving script persist a threshold in this shape — the project's own evaluation path computes recall at a target FPR and discards the threshold value.

An internal-consistency check the file fails. The comparison script does not recompute the pre-attack recalls; it copies them out of the same file, which makes them a free consistency test. On the matching $42{,}444$-row population the stored t_5 reproduces its published recall bitwise. The stored t_1 does not — it is short by exactly 15 rows. **The file is therefore internally inconsistent on the 1% side, not merely externally unverifiable.**

Q2 — staleness, and it is asymmetric. Re-deriving from the current corrected feature store shifts t_5 by 1.65×10^{-2} against the pre-correction vintage, and the stored t_5 reproduces its published recall only on the *pre-correction* store. t_1 is bit-identical across all four store vintages tested, so its inconsistency predates every feature-extraction fix and is a separate defect.

Q3 — what depends on it. The six per-attack recalls in Table 15 reproduce bitwise under the stored thresholds; re-deriving moves them between 0.05 and 3.5 points, all in one direction, with the largest movement on `sentence_reorder` at the 5% operating point, which this paper does not report. Both Class I effects survive in sign and order of magnitude. The cross-corpus transfer of §6 is unaffected: its primary block re-derives a RAID-native threshold and does not read this file, though per-domain and per-generator breakdowns computed elsewhere in that script do inherit it and are not reported here. Two consumers — the masked-versus-intact confound check and the attacker-capability ladder — were not re-settled, because both need live re-inference. **The ladder is reported in Appendix G and its tabular row carries this dependency**; the Binoculars and RoBERTa rows do not, so the monotonicity finding is unaffected.

G Attacker Capability Ladder

A note on floors first, since §7 relies on it: we treat a detector whose *pre-attack* recall@1%FPR is already at or below 15% as having no usable signal for an attack to destroy or spare, so its post-attack movement describes noise rather than robustness. Only `perplexity_gemma` (9.35%) falls below that screen; Fast-DetectGPT's 51.04% baseline sits far above it.

Table 17 gives the sweep behind §7: a single operator (paraphrase) regenerated by a size sweep of open instruction-tuned models on a $5{,}000$-document population, with a cross-family control. A sixth candidate (Falcon-7B-Instruct) was dropped for instruction-following failure rather than for its results.

Table 17: Recall@1%FPR under paraphrase by attacker model. The first four columns are the size-ordered Qwen2.5 sweep; Phi-3.5-mini is a cross-family control and is not part of that ordering, so the pattern column describes the Qwen sweep only. Source: `results/p4_robustness.md` §3.

Detector	Qwen2.5 0.5B	1.5B	7B	14B	Phi-3.5-mini (3.8B)	Pattern
`tabular`	72.6%	43.5%	45.1%	36.5%	37.7%	broadly declining across the Qwen sweep
`binoculars`	70.8%	58.6%	59.8%	35.1%	45.3%	cleanest decline; one inversion at 7B
`roberta`	99.7%	99.8%	98.8%	99.4%	99.9%	unmoved at every attacker size
`fastdetectgpt`	3.2%	2.4%	4.5%	1.2%	1.7%	floor-level under any non-lineage attacker

Regenerating the same paraphrase with models outside the scorer's lineage reverses the Class II effect entirely: Fast-DetectGPT's recall moves from 0.63–0.86 under same-lineage paraphrase to 0.012–0.045 across the five attackers above — a reversal large enough that no intermediate reading is ambiguous. **One clarification on provenance**: an earlier two-model diagnostic (Qwen2.5-7B-Instruct, Phi-3.5-mini-instruct), run before this ladder and on a separate population, put the same reversal at 0.025–0.157. It is the prerequisite investigation that prompted the ladder, not an independent measurement of it; the ladder supersedes it, and the range quoted in §7 and here is the ladder's. Reporting the attacker's lineage alongside the attack type would make the exposure visible without running anything.

Dependency note. The `tabular` row is computed against the operating-threshold file audited in Appendix F and inherits its unresolved provenance; it was not re-derived, because doing so requires live re-inference. The `binoculars`, `roberta` and `fastdetectgpt` rows do not read that file, so the monotonicity finding and the lineage reversal are unaffected. The Fast-DetectGPT row is the point of §7: the same detector that reads 0.63–0.86 recall under same-lineage paraphrase (Table 16)

reads 0.012–0.045 under any attacker outside its scorer's lineage. The apparent robustness is entirely an artifact of who generated the attack.

H MULTICLASS AND AUXILIARY RESULTS

Crossing corpora costs recall at 1% FPR $0.9587 \rightarrow 0.2277$, 95% CI [0.2127, 0.2391] (paired bootstrap, 2,000 resamples) — the largest single degradation in this evaluation, quoted in §6.

Table 18: Cross-corpus transfer, population-matched: both models scored on the identical 22,041-row RAID complete-case population, paired bootstrap, 2,000 resamples. The one place in this evaluation where the fractal family shows a significant advantage. Source: `data/fractalscope_check_raid_intersection.json`.

Model	n	AUROC	95% CI
`all` (85 features)	22,041	**0.8032**	[0.7975, 0.8089]
`all_minus_fractal` (31, same rows)	22,041	0.7792	[0.7730, 0.7851]
Paired bootstrap ΔAUROC $(85 - 31) = +\mathbf{0.0240}$, 95% CI [0.0222, 0.0259] — **excludes zero**			

Table 19: Auxiliary task results, for scope rather than for the fractal argument. Populations are named per row because n is not comparable across them. Sources: `data/p5_task{2,3,6}_*.json`.

Task	Population	n	Metric	Value
Generator family (Gemma vs. Mistral), 2-class	GAGLE complete-case	14,805	AUROC	0.9943 [0.9936, 0.9951]
same	same	14,805	TPR@1%FPR	0.9016 [0.8871, 0.9131]
Prompt type, 8-class, all features	GAGLE complete-case (85)	14,805	macro-F1	0.5816 [0.5747, 0.5885]
Prompt type, 8-class, `fractal_only`	GAGLE complete-case (54)	14,919	macro-F1	**0.1981** [0.1919, 0.2043]
Generator ID, 11-class	RAID complete-case	1,572	macro-F1	0.5441 [0.5164, 0.5684]
Edited vs. clean (single unpaired doc)	edit battery	7,439	AUROC	0.8705 [0.8602, 0.8820]
same	same	7,439	TPR@1%FPR	0.1420 [0.1049, 0.1913]

The same corpus supports the eight-class task at 0.58 macro-F1 with the full feature set and at 0.1981 with the fractal family alone, against a 0.125 chance floor. The two prompt-type arms differ in n by 114 rows because complete-case masking is computed over a different feature count; that cannot explain a gap of 0.38 macro-F1. The edited-versus-clean row deserves one note, because it was nearly reported at a much stronger level. A first-pass per-variant breakdown at the naive 0.5 threshold showed uniform 93.9–97.8% recall across all six operators and read as a generalised editedness signal. Checking the clean-side false-positive rate at that same threshold first gave 50.3% — half of clean documents were also called edited — so the table was measuring threshold permissiveness, not discrimination. Recomputed at the clean side's own 5%-FPR operating point, per-operator recall is 37.6–46.5%: still genuinely uniform, so the "not dominated by one operator" conclusion survives, at realistically modest numbers. Detectable in aggregate, weak at low FPR, from four scalar features.

H.1 P3 UNDER POPULATION CONTROL

The full run behind §4.2's population-control claim: a population-only control arm, a factorial-balance check on the post-gate population, and group-mean removal on the fractal columns, all against the same frozen prompt-type task (`full`, `lgbm`). The gate reproduces the frozen numbers to within their intervals (`all` 0.5837 against frozen 0.5816 [0.5747, 0.5885], inside the interval but not identical; `fractal_only` 0.1981 exact).

The population-only control. Given only generator, domain and temperature — no prompt type, no fractal features, no anything else — a classifier reaches macro-F1 0.0594, *below* the 8-class chance floor of 0.125. Population identity alone carries essentially no prompt-type signal: whatever the fractal-only arm's 0.1981 reflects, it is not leaking in through the population factors this control isolates.

Table 20: P3 population-control results. Source: `review_results/data/t3_population_control.json`.

Arm	n_{test}	macro-F1	Reading
Population-only (generator+domain+temp, no prompt type)	42,444	0.0594 [.0578, .0609]	*below* 0.125 chance
Fractal-only, before demeaning	14,919	0.1981 [.1919, .2043]	headline number
Fractal-only, after group-mean removal	14,919	0.2046 [.1986, .2111]	rises, does not fall
All-features, after fractal columns demeaned	14,805	0.5831 [.5759, .5899]	unchanged from 0.5816

Balance table. Across the 30 observed (generator, domain, temp) cells crossed with 8 prompt types (a 240-cell factorial by design), the maximum deviation of any cell's observed proportion from the design's uniform expectation is 0.0022 (0.22 percentage points, $\chi^2 = 413.3$ on 203 d.f.; the associated $p < 10^{-16}$ is a large-n artefact of 14,805 rows, not evidence of a meaningful imbalance — the raw deviation is the number that matters here). The factorial design survives the token-window and complete-case gates essentially intact.

Group-mean removal. Subtracting each of the 54 fractal columns' train-only (generator, domain, temp)-stratum mean and refitting is the direct test of whether 0.1981 is residual population leakage: if it were, demeaning should push the number toward chance. It does the opposite — $0.1981 \rightarrow 0.2046$, confidence intervals overlapping — while the all-features arm is unchanged ($0.5837 \rightarrow 0.5831$). Per the plan's own pre-registered interpretation rule (fixed before this table was produced): a result that holds near the un-demeaned value means the fractal family carries genuine document-level prompt-type signal, not population-structure leakage. **The number this analysis exposes as most uncertain is** 0.5816**, the all-features reference, not the fractal-only** 0.1981 **that the residual-confound objection targeted**: had the confound been real, 0.1981 would have fallen, and it did not.

Within-stratum distribution. Refitting separately inside all 30 cells (rather than pooling and controlling) gives all-features macro-F1 mean 0.567 (sd 0.032, range $[0.506, 0.666]$) and fractal-only mean 0.200 (sd 0.026, range $[0.147, 0.255]$) — both families sit well clear of chance in every one of the 30 cells, so the pooled 0.1981 is not an artefact of a few cells carrying the signal for the rest.

H.2 RAID GROUP-MEAN REPLACEMENT AND SUBTRACTION

The direct test of §6's population-mean account of the $+0.0240$ RAID transfer gain, run on the identical 22,041-row complete-case population as Table 18 (baseline AUROC reproduces exactly, 0.80321).

Table 21: RAID group-mean replacement and subtraction. (generator, domain) groups: the eight domains crossed with eleven generators and the human pool, 96 groups total. Source: `review_results/data/t5_raid_group_mean.json`.

Arm	AUROC	% of +0.0240 gap	Reading
`all` (85, reference)	0.8032	100%	baseline
`all_minus_fractal` (31, reference)	0.7792	0%	baseline
Replacement: fractal cols. → group mean	0.8149	**149%**	group mean alone *exceeds* the real gap
Subtraction: fractal cols. → $x - \bar{x}_k + \bar{x}$ (k the group)	0.7934	59%	most, not all, of the gap removed

Replacing each document's 54 fractal values with its own (generator, domain) group mean does not merely recover the gap, it *exceeds* it: group-mean-only fractal columns outperform the real per-document values on this transfer task. Subtracting the group mean (while adding back the global mean, so the frozen model still sees in-range inputs rather than values centred near zero it never trained on) retains 59% of the gap rather than collapsing it to zero. Read together: population-level (generator, domain) information does the majority of the work in the RAID transfer gain, with a real, non-trivial document-level residual left over — tilted toward the population-mean account, but not a clean "it is entirely the population mean" story.

I FEATURE INVENTORY

The 85 columns, as declared in `feature_store/p3_feature_manifest.json`. **Family A (fractal,** 54): $\widehat{S}$, $\widehat{H}$, log–log fit R^2 and bootstrap CI half-widths per scorer; per-scale log–log residuals at $\tau \in \{8, 16, 32, 48, 64\}$; intra-document windowed mean, variance, min, max and drift of both exponents; and the two cross-scorer divergences $|\widehat{S}_{\text{gem}} - \widehat{S}_{\text{mis}}|$ and $|\widehat{H}_{\text{gem}} - \widehat{H}_{\text{mis}}|$. **Family B (token-statistical,** 17): log-perplexity mean, variance, skewness and kurtosis per scorer; GLTR-style rank-bucket fractions (Gehrmann et al., 2019); exact next-token entropy; and a cross-scorer perplexity ratio. **Family C (discourse/style,** 14): sentence and paragraph length statistics, type–token ratio, bigram repetition, sentence-transition entropy, four discourse-marker rates, and MiniLM embedding dispersion, all computed on a per-document ratio-derived word budget so they cannot degenerate into length proxies.

Four of eight prompt types largely fail the Family-B alignment gate ($r \geq 0.98$) by design; the failure is recorded with an explicit `Alignment_Passed=False` flag and never imputed. Eleven provenance flags and six window-length columns are excluded from the feature matrix entirely, and a `provenance_only` arm prices that exclusion at AUROC 0.5655 (Appendix C).

J RELATED WORK, IN FULL

Long-range correlation in text is an old observation (Ebeling & Pöschel, 1993; Altmann et al., 2012), formalised for information-theoretic complexity by Alabdulmohsin et al. (2024) with $H \approx 0.7$ and extended to model output by Alabdulmohsin & Steiner (2025), whose narrow-band/wide-band contrast is our starting point. Parallel lines measure self-similarity by correlation dimension over next-token log-probability vectors (Du & Tanaka-Ishii, 2025), by scale-free property tests (Takahashi & Tanaka-Ishii, 2019; Meister & Cotterell, 2021), by spectral scaling of contextual embedding trajectories (Yang et al., 2026), and by the persistent-homology dimension of a document's token-embedding cloud (Tulchinskii et al., 2023; Wei et al., 2025). *We differ* in the question: not whether the scaling structure is present or informative in ensemble — we confirm it is — but whether the estimator recovering it has enough support on one document to say anything about that document. To our knowledge no prior work reports a reliability coefficient for these estimators at document length.

On the detection side, log-probability thresholds and rank visualisation (Gehrmann et al., 2019), curvature-based zero-shot scores (Mitchell et al., 2023; Bao et al., 2024), cross-model perplexity ratios (Hans et al., 2024) and fine-tuned encoders are the incumbents against which any additional feature family must be priced. West et al. (2025) argue that aggregating token-level measurements into a scalar discards positional information; our result is a complementary caution — retaining the whole scaling curve does not help either, because the curve is estimated on the same short support. *We differ* in treating the fractal family as an *additive* component priced against these incumbents rather than as a replacement, and in pricing it by permutation importance on held-out folds rather than by split gain.

Paraphrase defeats most detectors (Krishna et al., 2023), recursive paraphrase more so (Sadasivan et al., 2025), and prompt- or policy-optimised attackers more so again (Lu et al., 2024; Cheng et al., 2025); RAID (Dugan et al., 2024) is the standard shared benchmark for evaluating these claims across generators, domains and attacks. *We differ* in using the edit battery as a *structural probe* rather than as a leaderboard: the operators are partitioned a priori by what they act on, and the partition is used to predict the sign of a feature family's contribution, which is a different use of an attack cascade than measuring how much recall it removes.

Several recent evaluations find detector performance far below headline claims once the evaluation population is not the training population (Tufts et al., 2024; Dugan et al., 2024). *We differ* in supplying a mechanism: the reliability coefficient of §3 bears on *why* a particular representation does not transfer, and was taken before the transfer experiments were run rather than as an observation that it did not.

Finally, the operator partition of §5 is a general construction rather than a property of this feature family: for any representation defined as a functional of some structure in the input, an operator acting on that structure and one acting around it form a two-sided test. We would expect it to

separate positional from non-positional detectors (West et al., 2025; La Cava & Tagarelli, 2026) and, applied to a scoring model rather than a detector, to bear on the reversal-invariance property of the causal language-modelling objective (Sahasrabudhe, 2025).

K What Prior Work Says About Its Own Estimator

The two passages from Tulchinskii et al. (2023) referenced in §1, quoted in full because the framing of this paper depends on them. Their construction is a per-document scaling estimate: they compute the persistent-homology dimension "of each text sample, considering it as a separate manifold" by fitting a log–log linear regression over a sweep of subsample sizes, report a distribution-level separation (9.49 ± 1.01 for human text against ≈ 8 for generated), and then threshold the per-document estimate as a detector. Their first stated limitation is that the estimator "is stochastic in nature: PH dimensions of texts from the same generator vary widely, and the estimation algorithm is stochastic as well, which adds noise" — the document term (δ) and the estimator term (η) of Eq. 2 named in a single sentence, and separated in neither that work nor, to our knowledge, since. Observing that their misclassified outliers are disproportionately short, they further write that "PH dimension estimation is overall less correct on short texts" and "leave a more comprehensive analysis of such failure cases for future work." The split-half coefficient this paper reports is one way of running that analysis, and it is estimator-agnostic: it requires only that the quantity be recomputable on a subsequence.

Wei et al. (2025) confront the same problem at short length, trace it to a "local density peak" in the same log–log fit, and write the detection probability as $\Phi\big((\mu^h - \mu^m)/\sqrt{(\sigma^h)^2 + (\sigma^m)^2}\big)$. Their σ is the between-document standard deviation, which contains $\mathrm{Var}(\delta) + \mathrm{Var}(\eta)$ and cannot separate them; their remedy is to enlarge the support by inserting off-topic content before each document, which lowers σ from 1.49 to 0.79 on one subset and 1.54 to 1.00 on another while preserving the class gap.

L Extended Limitations

The symmetric edit battery, and why the requirement is a contribution rather than an omission. The asymmetry is a property of the evaluation design, not of the hypothesis, and it was identified by direct inspection of the perturbation pool ($5{,}000$ Mistral and $4{,}997$ Gemma source documents, zero human) rather than inferred after a null result. Stating it converts an unrunnable experiment into a runnable one, and it generalises beyond this project: the same leakage channel opens in any self-perturbation feature whose perturbed pool is drawn from a single class, which includes the natural way to build one from an existing adversarial-edit corpus. The cost of the fix is a generation-and-rescoring pass at roughly the scale of the existing cascade. The protocol, stated so the experiment is reproducible rather than merely named: draw a class-balanced sample spanning both human and machine documents; apply the same operator bank to both sides; rescore both through the identical Family A–C pipeline; and compute the drift feature symmetrically, so that non-nullity carries no label information.

No architectural account of the exponents. We identify one architecture-level mechanism (scorer–attacker lineage, §7) and explicitly decline the other. Establishing *why* a given decoder produces trajectories with a given H would require the exponents measured against controlled interventions on the generation process — temperature, decoding strategy, context length, attention range — with the scorer held fixed. GAGLE varies temperature across three settings, which is one such axis; we did not analyse the exponents against it, and we do not substitute a plausible story for the measurement.

Human revision has no genuine coverage. The battery's seventh operator was excluded as a no-op (89.2% near-identical outputs, cosine > 0.98), so the paper's robustness claims cover machine-executed edits only. How these representations behave under real human revision of machine drafts — the condition most detection deployments actually face — is untested here.

Detection granularity. All results are document-level. Window statistics enter only as columns inside Family A; a sliding-window classifier and a corpus-level pooled decision, both of which the estimand/support account predicts should behave differently, were never built. Given §3.2, corpus-level aggregation is the regime where these estimators are on firmest ground, and we have not measured how many documents of pooling it takes to recover usable signal.

Corpus-specific confounds. Human documents are far longer than generated ones here (mean 691 against $\approx$ 318 words) and human prevalence is 2.234%; we price length explicitly (§4) and report every metric at fixed low-FPR operating points with class weighting and calibration at the validation split's own prevalence — *not* at corpus prevalence and not at any deployment prevalence, which is unknown and differs from both. A deployer must recalibrate. The token-window gate drops 7,397 machine rows whose median length is 128 words against 631 for retained rows, concentrated in two free-form prompt types (`excerpt` 9.62% and `continue_pt` 8.37% drop rates, together 73% of all drops); those are the same prompt types that are hardest to detect in §4.2, so the dropped and the missed populations point at the same place by two different mechanisms.

Matched-length bands, measured. A sharper version of the length confound than the generic warning above: GAGLE's human-side documents are gated to *exactly* 400 usable tokens with zero variance (sd $= 0$ across all 81,238 human doc/scorer/feature rows), while LLM documents range 320–463. Consequently the bands 320–360 and 401–464 are LLM-only by construction, and a genuine human/machine length-matched comparison is achievable only inside the narrow 361–400 band, itself dominated by the fixed human value of 400 — reported as the finding and the limitation, rather than forcing an unsupported match.

Table 22: Reliability (ρ_{pooled}) by usable-token band. n varies sharply by band (2k–11k in the edge bands, 180k+ in the middle band). Source: `review_results/data/t7_matched_length.json`.

Band	`gemma_S`	`gemma_H`	`mistral_S`	`mistral_H`
320–360 (LLM-only)	0.091	0.412	0.003	0.476
361–400 (mixed, human-heavy)	0.055	0.582	0.045	0.546
401–464 (LLM-only)	0.109	0.478	0.141	0.493

$\widehat{S}$ reliability is consistently higher in the longer 401–464 band than the 361–400 band for both scorers — a small effect, but the right sign for the support mechanism §3.2 proposes, visible even inside this narrow token range. **Across all three bands the trend is not monotone**, and we report that rather than only the supporting pair: only `mistral_S` rises at every step ($0.003 \to 0.045 \to 0.141$), while `gemma_S` dips in the middle band and both $\widehat{H}$ columns peak there. The middle band is also the best-powered one (180k+ rows against 2–11k in the edge bands), so we do not read these bands as a clean support gradient; a 144-token range is too narrow to resolve one, which is why the support question is settled by the $N^\star$ construction of Appendix B.4 instead.

Multiplicity, stated rather than assumed away. Beyond the corrected 40-test grid of §4.1, a separate 65-comparison paired-significance run against external baselines applied no multiplicity correction, so ≈ 3 false positives are expected at $\alpha = 0.05$ by chance alone. Two of the 65 are individually non-significant, and both sit on comparatively small human populations; neither should be read as a parity result.

Protocol fidelity, and an exhaustive column-level audit. We use the same estimator family as the source publication but have not independently cross-checked every numeric protocol choice against its published code — our window constants (64/464/320) are the shipped code's, and differ from the plan documents' own description (400/400), a discrepancy we resolve in favour of the artifact. To rule out silent feature-extraction failures rather than assume their absence, we audited every column of the 85-feature matrix against its edge cases directly; the audit surfaced two defects, both recorded here rather than patched without a trace: a degenerate Hölder branch returning an indistinguishable NaN on $\approx 0.05\%$ of documents in both corpora, now carrying a distinguishing reason code (that tail is the ε calibration of §2 failing on an individual document whose z-scored trajectory drifts too far for any displacement to land inside ε at any tested lag: p_ε is then exactly zero at every τ and no slope

exists. The three published sources for this estimator use three different ε — 10^{-2}, 5×10^{-2} and 10^{-1} — on documents of different length, and the reference implementation relies on the z-scoring of Eq. 1 to make a single fixed ε portable across documents at all, which is why ours is calibrated against this corpus's own length-to-scale ratio rather than adopted from any one of them); and an ASCII-only word-tokenisation regex that mis-counted 13 of the 85 columns on any non-ASCII text, affecting 99.98% of RAID's homoglyph-attack rows (median -44.75% word count). All feature stores were rebuilt and every dependent result regenerated. Every number in this paper is post-correction *with one identified exception*: the pre-attack recall@5%FPR of Table 16 descends from a threshold that reproduces only against the pre-correction store (Appendix F). It is a copy-through in a row already marked as one, and no claim in this paper depends on it.

Zero-width-space rows. The matched RAID comparison covers $22{,}041$ of $68{,}212$ rows; the excluded remainder is disproportionately affected by the alignment and window gates of §3, and one attack interacts with them in particular — zero-width-space insertion drops 84.3% of its rows under complete-case, against 64.7% for other RAID attacks, because the fractal scale columns genuinely need more real tokens than a whitespace-inflated document supplies. For the surviving zero-width-space rows the feature vector is a deliberate hybrid — style features computed on stripped text, fractal and perplexity features from a scoring pass that saw the true inserted stream — so part of the vector already defeats the attack and part still sees it. We therefore do not read RAID recall as a single coherent judgment of that attack's effect. The complete-case exclusion is not attack-neutral: among excluded rows, zero-width-space accounts for 24.3% of the attack composition against 9.5% among included rows (all other attacks are within ±3 points of parity between included and excluded), so complete-case AUROC figures reported elsewhere in this paper are, if anything, biased toward the attacks the detector handles best.

Table 23: Complete-case AUROC by attack, domain and generator (baseline complete-case AUROC $= 0.803$). Source: `review_results/data/t9_raid_complete_case.json`.

By attack	n	AUROC
zero_width_space (weakest)	2085	0.650
homoglyph	4216	0.711
paraphrase	1686	0.783
upper_lower	2207	0.836
article_deletion	2148	0.845
perplexity_misspelling	2257	0.850
alternative_spelling	2174	0.861
none (no attack)	2171	0.863
whitespace (strongest)	3097	0.881

By domain	n	AUROC
reddit (weakest)	1652	0.592
recipes	3510	0.660
poetry	2705	0.718
abstracts	1401	0.724
wiki	2731	0.814
books	3065	0.830
reviews	3138	0.881
news (strongest)	3839	0.930

The generator-versus-human breakdown (11 generators, $n = 11{,}757$–$12{,}840$ each against the shared human pool) ranges from AUROC $= 0.697$ (`mpt-chat`) to 0.899 (`chatgpt`); the detector's single weakest generator-attack combination overall is `mpt-chat` under zero-width-space, which is also the combination most disproportionately removed by complete-case filtering. Read together with the exclusion-composition result above, the complete-case AUROC numbers elsewhere in this paper should be read as an upper bound on robustness to zero-width-space specifically, not a representative average over it.

Single scorer pair, single corpus family, English only. Both scoring models are open-weight base models of 2B and 7B parameters; the generators are the same two models. Cross-lingual behaviour is untested, and was deferred deliberately rather than overlooked — log-perplexity behaviour for these scorers is unverified across languages, so adding multilingual text would open a new validity question rather than enlarge the existing one.

Estimator invariance, part 2: persistent homology on a structurally distinct representation. Part 1 of this argument (Appendix B.5) rules out an R/S-specific artifact using two estimators that still measure the same scale exponent on the same surprisal sequence. This pilot removes that shared assumption entirely: persistent homology does not look at sequential variance at all — it takes a cloud of points (here, overlapping windows of a document's MiniLM embedding trajectory) with no built-in notion of order between them, connects nearby points into simplices (edges, triangles, and

their higher-dimensional analogues) at a growing distance threshold, and tracks which topological features — connected components (H_0) and loops (H_1) — persist across that threshold sweep. If the short-support ceiling recurred here too, on a representation built from geometric shape rather than from ordered fluctuation, that would be evidence the ceiling is a property of the estimand at short support in general, not an artifact of scale-exponent estimators specifically. §8 reports a best genuine value of 0.41 for this pilot on overlapping-window MiniLM embeddings; this paragraph gives the full geometry sweep, in the interest of the same transparency this paper applies to its own primary estimators. Windows of the embedding trajectory are swept over Vietoris–Rips persistent homology, summarised by two families of features: H_0 (connected-component) and H_1 (loop) statistics. H_0 total persistence is a mathematical identity of this pipeline (it equals a monotone function of window count by construction, independent of any signal in the text) and is excluded as a non-finding rather than reported. The sweep also establishes what counts as a genuine measurement here: the two halves must be built from independent token content for a reliability estimate to mean anything. At window$=40$/stride$=20$ and window$=40$/stride$=8$, the two halves share 90.6% and 96.3% of their token content by Jaccard overlap respectively — the halves are not independent at either geometry, so the high reliability both appear to show is largely a restatement of shared content, not evidence of the representation's stability; both are kept on disk rather than deleted, consistent with this paper's practice elsewhere in this appendix, but neither is reported as a finding. Restricting the sweep to zero-overlap geometries (window=stride, so consecutive windows share no tokens) gives the results in Table 24, of which window$=15$/stride$=15$, H_1 total persistence, at 0.41, is the best genuine value found.

Table 24: Persistent-homology pilot, zero-overlap geometries only. Source: `review_results/data/t10_pilot_geometry_comparison.json`.

Geometry (window/stride)	Feature	ρ_{SB}
w=40, s=40	H_1 total persistence	0.29
w=15, s=15	H_1 total persistence	0.41
w=10, s=10	H_1 total persistence	0.33

None of the zero-overlap geometries clears the 0.60 bar, and none reaches even $\widehat{H}$'s own 0.576 at GAGLE's support. We read this as a negative, generalizing result rather than a failure of this pilot: a representation with no fractal machinery at all, built on overlapping-window embeddings rather than token-level scale features, shows the same short-support reliability ceiling this paper documents for $\widehat{S}$. This is one pilot on one corpus and is not a substitute for the primary analysis; it is reported because it bears on whether the reliability ceiling is a property of this paper's specific estimators or of short language-model text more broadly, and the pilot's answer, so far as it goes, favours the latter.

M FIGURES

The reliability-under-alternative-splits and reliability-versus-support figures referenced from the main text are in Appendix B (Figures 2 and 3); Figure 1, the paper's summary figure, now appears in the main text (§3). The reliability gate, the robustness battery and the transfer dissociation are also available as standalone diagrams (the `fig_p2`, `fig_p4` and `fig_p5` sources under `supplementary_figures/`), which compile against the shared Okabe–Ito preamble already inlined in this document's header, and are included in the supplementary material.